Çelik Yapılar I

Steel Structure I

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Syllabus

Steel Structure I		
Syllabus		
INS3531 Steel Structure I course plan is scheduled as follows;		
Weeks	Chapter	Exams
1	Introduction, Material Properties, Loads and Load Combinations in Steel Structures	
2	Desing of Tension Members	
3	Desing of Tension Members	
4	Design of Compression Members	
5	Design of Compression Members	
6	Design of Flexural Bending Members	
7	Design of Flexural Bending Members	
8	Midterm 1	1st Midterm Exam
9	Design of Beam-column Members	
10	Design of Beam-column Members	
11	Simple Connections (Bolted and Welded)	
12	Simple Connections (Bolted and Welded)	
13	Eccentric Connections (Bolted and Welded), Mid-Term Evaluation	2nd Midterm Exam
14	Eccentric Connections (Bolted and Welded)	
15	Final	

Ders Öğrenim Çıktıları

- Öğrenci, çelik yapı elemanlarının tasarım yüklerini hesaplayabilir.
- Öğrenci, çelik yapı elemanlarının hesaplarını yapabilir.
- Öğrenci, çelik yapılarda kullanılan birleşim ve birleştirme vasıtalarının hesaplarını yapabilir.
- Öğrenci, çelik yapılardaki stabilite problemlerini anlayarak boyutlandırma yapabilir.
- Öğrenci, güncel şartnameleri takip etmenin önemini fark ederek güncel yöntemleri anlayıp yorumlama becerisi kazanabilir

TÜRKİYE YÜKSEKÖĞRETİM YETERLİLİKLER ÇERÇEVESİ (TYYÇ) VE PROGRAM ÇIKTISI (PÇ) İLİŞKİ MATRİSİ

	BİLGİ		BECERİLER		YETKİNLİKLER			
	Kuramsal	Uygulamalı	Kavramsal/Bilişsel	Uygulamalı	Bağımsız Çalışabilme ve Sorumluluk Alabilme yetkinliği	Öğrenme Yetkinliği	İletişim ve Sosyal Yetkinlik	Alana Özgü Yettkinlik
PÇ-1	X	X	X	X				X
PÇ-2			X	X	X			X
PÇ-3	X	X	X	X	X	X		X
PÇ-4			X		X		X	
PÇ-5	X	X	X	X				
PÇ-6								X
PÇ-7							X	
PÇ-8		X						X
PÇ-9				X	X	X		X
PÇ-10								X
PÇ-11	X		X				X	

Ders & Program Çıktıları Matrisi

Ders & Program Çıktıları Matrisi

		Program Çıktıları										
Kodu	Ders Adı	1	2	3	4	5	6	7	8	9	10	11
INS3541	Yapı Statiği 2	5	5	4	4	3	4	4	5	4	3	4
INS3211	Hidrolik	5	5	5	5	5	1	5	5	3	3	5
INS3121	Zemin Mekaniği	5	5	4	3	5	5	3	4	4	4	4
INS3531	Çelik Yapılar 1	5	3	4	2	5	4	1	3	3	2	4

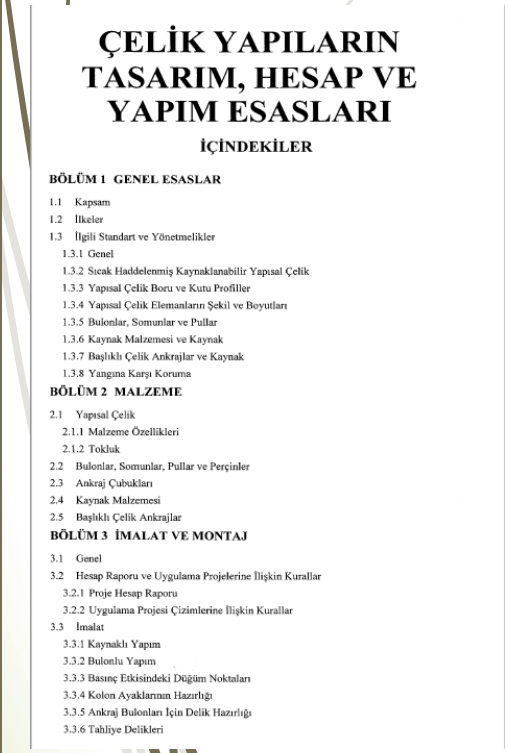
Çelik yapılar kürsüsü ders program çıktıları

Destekelenen PÇ Sayısı	DERS KODU	DERSLER	ANABİLİM DALI	PÇ 1-2	PÇ 1-3	PÇ 3-1	PÇ 6-1	PÇ 9-2
4	INS3531	Çelik Yapılar 1	YAPI	GD	GD		ÖD	ÖD
4	INS4541	Çelik Yapılar 2	YAPI	GD	GD		ÖD	GD
4	INS4522	Çelik Yapıların Plastik Hesabı	YAPI	GD	GD		ÖD	GD
2	INS4631	Çelik Yapı Tasarımı	YAPI			ÖD	ÖD	

KANITLAR	
1. VİZE	1V
2. VİZE	2V
PROJE	PR
ÖDEV	ÖD
DİĞER	D
FİNAL	FN
GENEL DEĞERLENDİRME	GD

Standartlar/Standards, Codes or Specifications

Türkiye Çelik Yapıların Tasarım, Hesap ve Yapım Esasları 2016



ÇELİK YAPILARIN TASARIM, HESAP VE YAPIM ESASLARI

İÇİNDEKİLER

BÖLÜM 1 GENEL ESASLAR

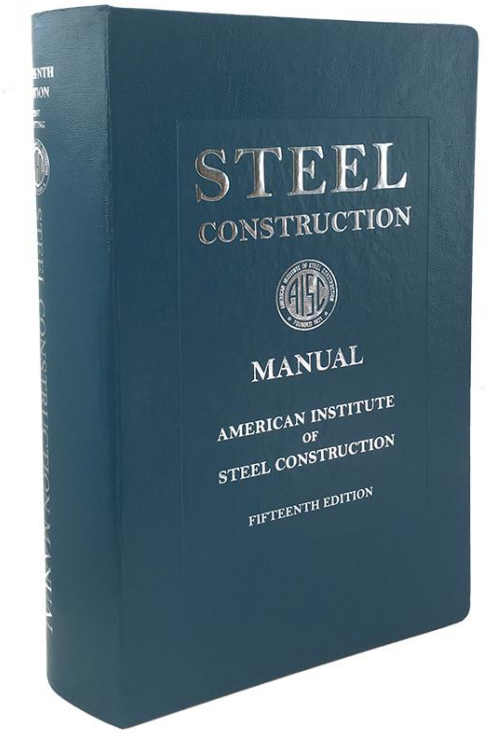
1.1 Kapsam
1.2 İlkeler
1.3 İlgili Standart ve Yönetmelikler
1.3.1 Genel
1.3.2 Sıcak Haddelenmiş Kaynaklanabilir Yapısal Çelik
1.3.3 Yapısal Çelik Boru ve Kutu Profiller
1.3.4 Yapısal Çelik Elemanların Şekil ve Boyutları
1.3.5 Bulonlar, Somunlar ve Pullar
1.3.6 Kaynak Malzemesi ve Kaynak
1.3.7 Başlıklı Çelik Ankrajlar ve Kaynak
1.3.8 Yangına Karşı Koruma

BÖLÜM 2 MALZEME

2.1 Yapısal Çelik
2.1.1 Malzeme Özellikleri
2.1.2 Tokluk
2.2 Bulonlar, Somunlar, Pullar ve Perçinler
2.3 Ankraj Çubukları
2.4 Kaynak Malzemesi
2.5 Başlıklı Çelik Ankrajlar

BÖLÜM 3 İMALAT VE MONTAJ

3.1 Genel
3.2 Hesap Raporu ve Uygulama Projelerine İlişkin Kurallar
3.2.1 Proje Hesap Raporu
3.2.2 Uygulama Projesi Çizimlerine İlişkin Kurallar
3.3 İmalat
3.3.1 Kaynaklı Yapım
3.3.2 Bulonlu Yapım
3.3.3 Basınç Etkisindeki Düğüm Noktaları
3.3.4 Kolon Ayaklarının Hazırlığı
3.3.5 Ankraj Bulonları İçin Delik Hazırlığı
3.3.6 Tahliye Delikleri

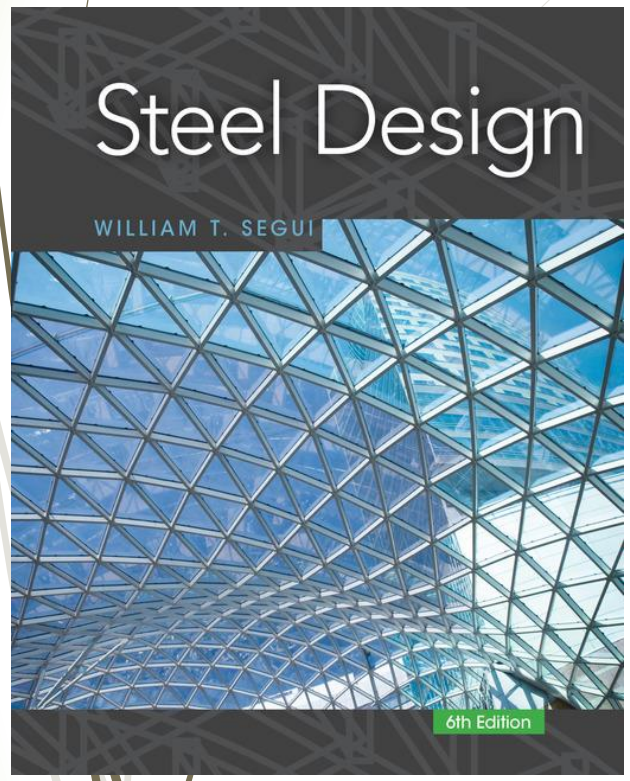


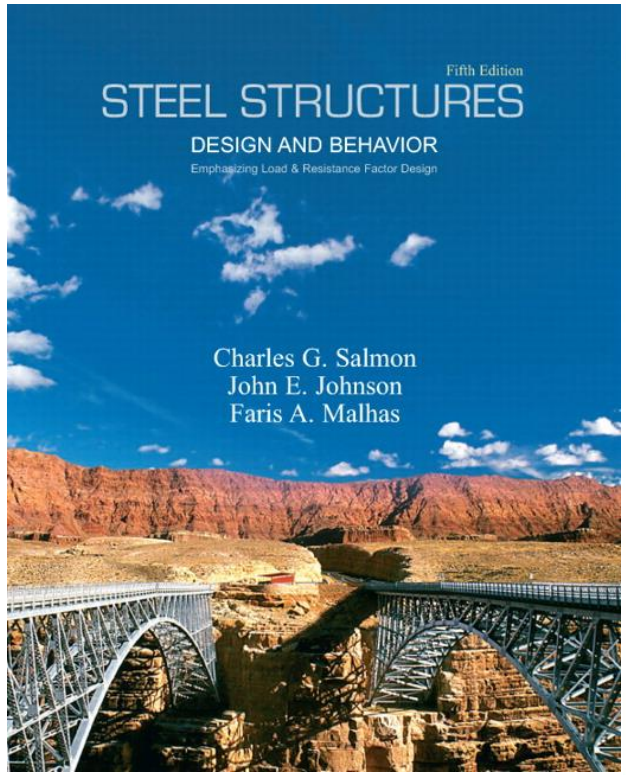


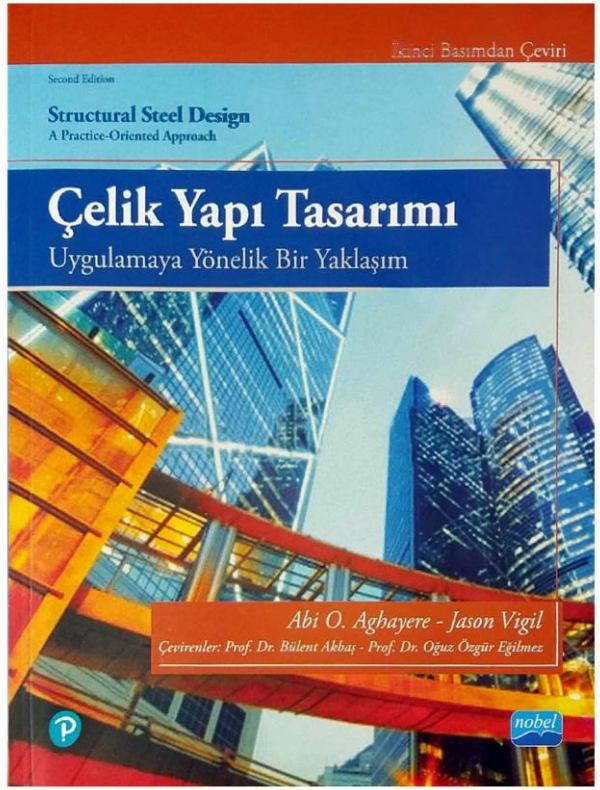
AASHTO
AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
1914

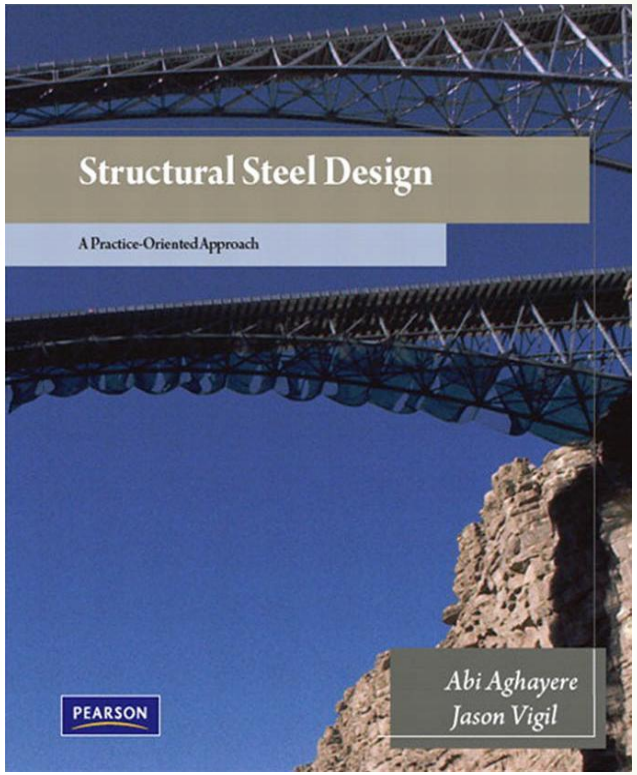
AISC
AMERICAN INSTITUTE OF STEEL CONSTRUCTION INC.
FOUNDED 1921

Kitaplar / Books









Youtube Videos

- Introduction to Basic Steel Design, https://www.youtube.com/watch?v=OoSAgbPhnzc
- Structural Steel Design Course, https://www.youtube.com/watch?v=zwXuF_3n_e0&list=PLCV9OyAY5K-VvmJjj0kB3ZY8ycAynOoP2
- Steel Fabrication : A Virtual, Detailed Tour of the Steel Fabrication Process, https://www.youtube.com/watch?v=6CFDIBILmZ8
- Rules of Thumb for Steel Design, https://www.youtube.com/watch?v=TM25D8RGiyk
- Steel Material Properties, https://www.youtube.com/watch?v=loRrbAjhZBw
- Fastener Fundamentals, https://www.youtube.com/watch?v=rt6D-p5BgO8
- High Strength Bolting: The Basics, https://www.youtube.com/watch?v=PmXMTt_JPb8
- Weld Details: The Good, The Bad and The Ugly, https://www.youtube.com/watch?v=mRJp75pN0Ho
- Weld Inspection: What Matters and What Doesn't, https://www.youtube.com/watch?v=jhxRagJo644
- Column Design: Past, Present, and Future, https://www.youtube.com/watch?v=L62qt8KcTcI
- What Engineers Need to Know about Steel Erection, https://www.youtube.com/watch?v=hrRyM4V89oU
- Introduction to the Steel Construction Process: The Team Behind the Building, https://www.youtube.com/watch?v=zZ5P-alBQaI
- Steel Design After College, https://www.youtube.com/watch?v=BJffavTtBz8

Notes

1. Students may attend to **midterms** and **final exam** with only "Çelik Yapıların Tasarım, Hesap ve Yapım Esaslarına Dair Yönetmelik (Turkish Steel Code 2016)" or AISC Steel Standard AISC 360-16. **Other course materials are not allowed.**
2. Final score of the student is determined as the weighted average of midterm exams (%60), and final exam (%40).
3. Lower bound for DC is 40.
4. Attendance will be taken into consideration for both theoretical and practical lectures. At the end of term, students with the attendance lower than %70 will get F0.

Notes

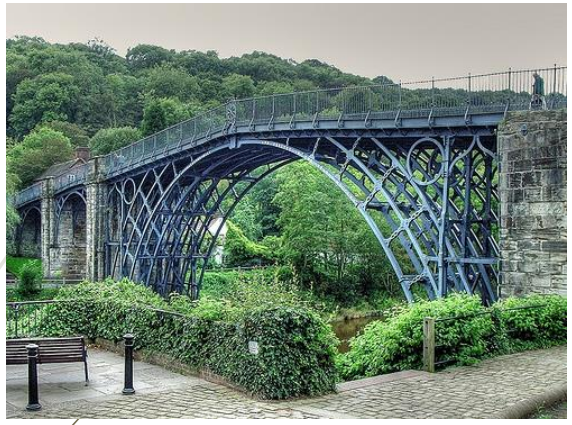
5. Exam duration for both final and midterm exams is estimated as 110 minutes.
6. Students are expected to attend the lectures of their own groups which they are registered. Transitions between groups are not allowed.
7. Recommended resources for this course;

- Akbaş, B.,Eğilmez, O.Ö., Çelik Yapı Tasarımı: Uygulamaya Yönelik Bir Yaklaşım (Structural Steel Design: A Practive-Oriented Approach), Nobel Yayınevi, 2.Basımdan Çeviri
- W. Segui, "Steel Design", 5th Ed., Cengage Learning, 2011
- C.G. Salmon, J.E. Johnson, F.A. Malhas, "Steel Structures: Design and Behavior Emphasizing LRFD", 5th. Ed., Pearson, 2009
- Çevre ve Sehircilik Bakanlıgı, "Çelik Yapıların Tasarım, Hesap ve Yapım Esaslarına Dair Yönetmelik", 2016

İçerik /Content

- Çelik yapıların tarihçesi /History of Steel Structures
- Çelik malzeme özellikleri /Material Properties of Steel
- Çekme Çubukları /Tension Members
- Basınç çubukları /Compression Members
- Kiriş Elemanları /Beams
- Kolon kiriş elamanı /Beam-columns
- Birleşim elemanları ve hesabı /Fasteners and its design

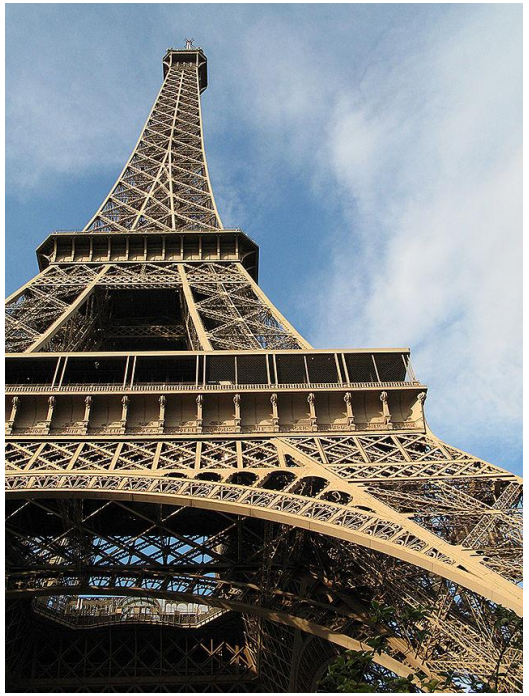
Çelik Yapılar tarihçesi /History of Steel Structures



Coalbrookdale köprüsü (eng 1778)



Bethlehem çelik fabrikası (usa 1857)



Eifel kulesi(fra 1860)

Çelik yüksek yapılar /High-rising (High Rise) Steel Buildings



Name	Taipei 101	Shanghai World Financial Center	Petronas Towers	Willis Tower
City	Taipei TpCi	Shanghai SH	Kuala Lumpur KL	Chicago IL
Country	Taiwan	China	Malaysia	United States
Illustrator	Someformofhuman	Ka777	Someformofhuman	Supertall
Status	Built	Built	Built	Built
Built	2004	2008	1998	1974
Floors	101	101	88	108
Use	Mixed use	Office	Office	Office
Antenna				527.3 m
Spire	508 m		452 m	
Roof	448 m	492 m		442.3 m

Çelik Yapılar tarihçesi /History of Steel Structures







Haliç köprüsü (Tur 1875)



Boğaziçi
Köprüsü
(Tur 1973)

Pros and Cons

Independence from weather conditions

Low space requirement for building and building equipment

Short construction time and low construction cost

High strength building elements



High degree of industrial pre-production

Passing through large span with low cross-sectional dimension of elements

High fire safety in composite structures

High precision sizing, freedom of installation

Ability to provide simple solutions to subsequent changes in intended use

Yapıya etkiyen yükler / Loads

Bir yapıyı etkileyen yükler, esas yükler ve ilave yükler olarak ikiye ayrılır: *(Loads affecting a structure are classified as principal loads and additional loads.)*

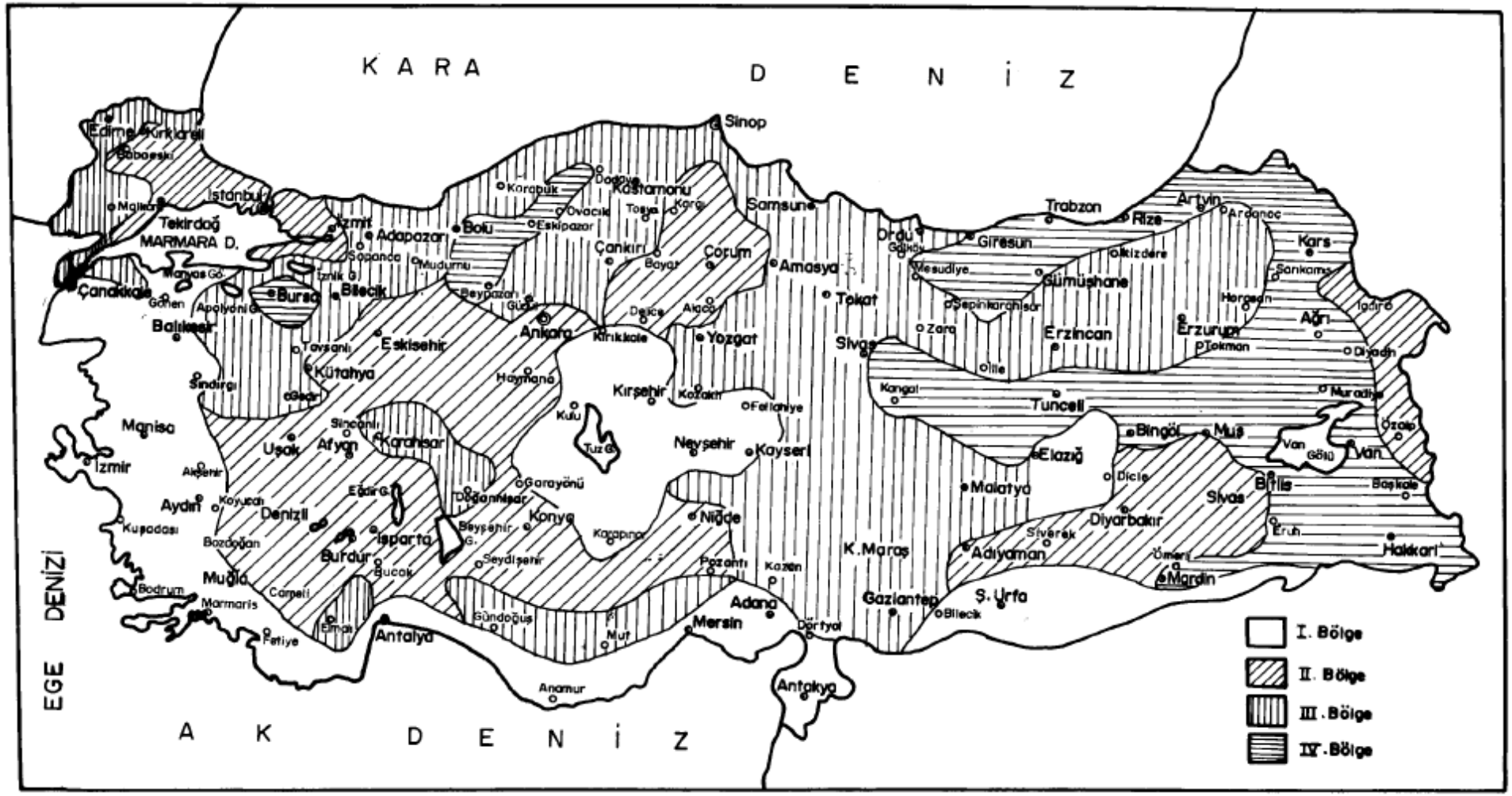
- Esas yükler: Öz yükleri, munzam ve hareketli yükleri, kar, makinelerin kütle kuvvetlerini kapsar. *(Base Loads: It includes self-loads, live loads, the mass forces of machines, and snow)*.

- İlave yükler: Rüzgar etkisini, deprem etkisini, fren kuvvetlerinin yanal bileşeni (kreynlerde) seyrek olarak montaj ve tamir işlerinde kullanılan işler haldeki (kreynlerde) ısı etkilerini (işletmeye bağlı ve atmosferik) kapsar. *(Additional Loads: It includes wind effect, earthquake effect, lateral component of brake forces (in cranes) and heat effects (operational and atmospheric) in working conditions (in cranes) which are rarely used in assembly and repair works.)*

Yapıya etkiyen yükler / Loads

- Yapı sistemlerinin tasarımında esas alınan karakteristik yük değerleri, TS 498 e uygun olarak belirlenecektir.
- Kar yükleri için TS EN 1991 – 1 – 3
- Rüzgar yükleri için TS EN 1991 – 1 – 4 te verilen koşullar gözönüne alınacaktır.
- Deprem etkisi, *E* nin ayrıntılı tanımı için Türkiye Bina Deprem Yönetmeliği 2018 koşulları esas alınacaktır.
- Rüzgar hızının temel değeri, *V*b,o = 28 m/sn (100 km/sa) den ve binanın ana taşıyıcı sistemine, dış cephe kaplamalarına ve rüzgara maruz yapısal ve yapısal olmayan elemanlarına etkiyen karakteristik rüzgar yükleri 0.5 kN/m2 den az olmayacaktır.

Kar Yükü / Snow Load


KARA DENİZ
EGE DENİZİ
AK DENİZ
Sinop
Samsun
Trabzon
Rize
Artvin
Ardanuç
Giresun
Ordu
Gümüşhane
Kars
Sarıkamış
Erzincan
Erzurum
Tekman
Ağrı
Iğdır
Amasya
Tokat
Çorum
Çankırı
Sivas
Yozgat
Ankara
Kırıkkale
Kırşehir
Nevşehir
Kayseri
Niğde
Konya
Aksaray
Karaman
Eskişehir
Afyon
Uşak
Kütahya
Bursa
Bilecik
Balıkesir
Çanakkale
İstanbul
Tekirdağ
Edirne
Kırklareli
İzmir
Manisa
Aydın
Denizli
Muğla
Bodrum
Marmaris
Isparta
Burdur
Antalya
Mersin
Adana
Antakya
Osmaniye
Gaziantep
Şanlıurfa
Adıyaman
Kahramanmaraş
Malatya
Elazığ
Bingöl
Muş
Bitlis
Van
Hakkari
Diyarbakır
Mardin
Batman
Siirt
Şırnak
Tunceli
Kangal
Zara
Ilıca
Divriği
Hafik
Suşehri
Gölköy
Mesudiye
Ünye
Fatsa
Çarşamba
Samsun
Havza
Vezirköprü
Bafra
Terme
Ayancık
Boyabat
Durağan
Gerze
Ayvacık
Karapürçek
Çerkeş
Ilgaz
Kastamonu
Tosya
Çankırı
Çorum
Osmancık
Sungurlu
Mudurnu
Göynük
Bolu
Gerede
Düzce
Akçakoca
Ereğli
Zonguldak
Karabük
Safranbolu
Yenice
Devrek
Çaycuma
Bartın
Amasra
Cide
Eskipazar
Ovacık
Taşköprü
Kargı
Osmancık
Sungurlu
Çerkeş
Ilgaz
Kurşunlu
Orta
Çankırı
Şabanözü
Yapraklı
Çandır
Kızılcahamam
Çamlıdere
Nallıhan
Beypazarı
Güdül
Ayaş
Polatlı
Haymana
Kırıkkale
Delice
Akdağmadeni
Yerköy
Sorgun
Yozgat
Çekerek
Boğazlıyan
Sarıkaya
Şefaatli
Sorgun
Akdağmadeni
Kadışehri
Yerköy
Çayıralan
Saraykent
Sarıkaya
Yıldızeli
Gemerek
Şarkışla
Divriği
Kangal
Hafik
Zara
İmranlı
Gürün
Suşehri
Ulaş
Koyulhisar
Refahiye
Tercan
Üzümlü
Kemaliye
Çayırlı
Otlu
Kemah
Tekman
Karayazı
Köprüköy
Pasinler
Hınıs
Tekman
Karaçoban
Karlıova
Varto
Hınıs
Tekman
Karayazı
Köprüköy
Pasinler
Aşkale
Çat
Horasan
Narman
Oltu
Olur
Şenkaya
Tortum
Karayazı
Köprüköy
Pasinler
Aşkale
Çat
Horasan
Narman
Oltu
Olur
Şenkaya
Tortum
Uzundere
Pazaryolu
Ispir
Yusufeli
Artvin
Borçka
Hopa
Şavşat
Ardanuç
Arhavi
Kemalpaşa
Görele
Tirebolu
Espiye
Eynesil
Bulancak
Keşap
Şebinkarahisar
Alucra
Doğankent
Piraziz
Yağlıdere
Çamoluk
Çanakçı
Dereli
Espiye
Eynesil
Bulancak
Keşap
Şebinkarahisar
Alucra
Doğankent
Piraziz
Yağlıdere
Çamoluk
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Tirebolu
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Bafra
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Ayancık
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Durağan
Gerze
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Karapürçek
Çerkeş
Ilgaz
Kastamonu
Tosya
Çankırı
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Mudurnu
Göynük
Bolu
Gerede
Düzce
Akçakoca
Ereğli
Zonguldak
Karabük
Safranbolu
Yenice
Devrek
Çaycuma
Bartın
Amasra
Cide
Eskipazar
Ovacık
Taşköprü
Kargı
Osmancık
Sungurlu
Çerkeş
Ilgaz
Kurşunlu
Orta
Çankırı
Şabanözü
Yapraklı
Çandır
Kızılcahamam
Çamlıdere
Nallıhan
Beypazarı
Güdül
Ayaş
Polatlı
Haymana
Kırıkkale
Delice
Akdağmadeni
Yerköy
Sorgun
Yozgat
Çekerek
Boğazlıyan
Sarıkaya
Şefaatli
Sorgun
Akdağmadeni
Kadışehri
Yerköy
Çayıralan
Saraykent
Sarıkaya
Yıldızeli
Gemerek
Şarkışla
Divriği
Kangal
Hafik
Zara
İmranlı
Gürün
Suşehri
Ulaş
Koyulhisar
Refahiye
Tercan
Üzümlü
Kemaliye
Çayırlı
Otlu
Kemah
Tekman
Karayazı
Köprüköy
Pasinler
Hınıs
Tekman
Karaçoban
Karlıova
Varto
Hınıs
Tekman
Karayazı
Köprüköy
Pasinler
Aşkale
Çat
Horasan
Narman
Oltu
Olur
Şenkaya
Tortum
Karayazı
Köprüköy
Pasinler
Aşkale
Çat
Horasan
Narman
Oltu
Olur
Şenkaya
Tortum
Uzundere
Pazaryolu
Ispir
Yusufeli
Artvin
Borçka
Hopa
Şavşat
Ardanuç
Arhavi
Kemalpaşa
Görele
Tirebolu

Kar Yükü / Snow Load

Çizelge MA.1 - Karakteristik zemin kar yükü (s_k) değerleri kN/m^2 (*)

	1	2	3	4	5
1	Yapı yerinin denizden yüksekliği	BÖLGELER			
	m	I	II	III	IV
	≤ 200	0,75	0,75	0,75	0,75
2	300	0,75	0,75	0,75	0,80
	400	0,75	0,75	0,75	0,80
	500	0,75	0,75	0,75	0,85
3	600	0,75	0,75	0,80	0,90
	700	0,75	0,75	0,85	0,95
	800	0,80	0,85	1,25	1,40
4	900	0,80	0,95	1,30	1,50
	1000	0,80	1,05	1,35	1,60
5	> 1000	1000 m'ye tekabül eden değerler, 1500 m'ye kadar % 10, 1500 m'den yukarı yüksekliklerde % 15 artırılır.			
* Kar yağmayan yerlerde kar yükü hesap değeri sıfır alınır.					

TS EN 1991-1-3 Madde 5.2 Yük düzenlemeleri

$$s = \mu i \; Ce \; Ct \; sk$$

Burada;

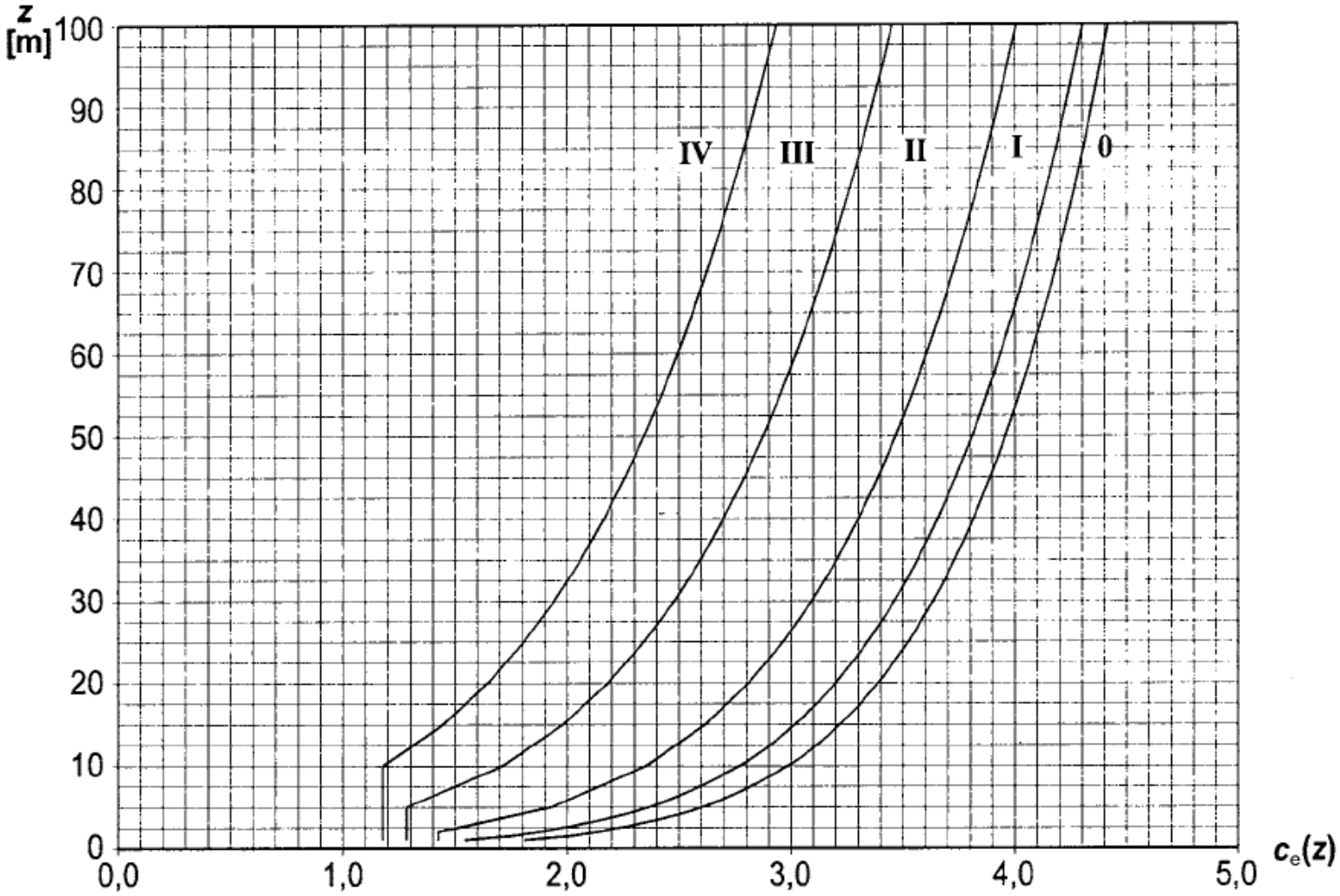
μi: Kar yükü şekil katsayısı (Madde 5.3 ve Ek B), is the snow load shape coefficient (see Section 5.3 and Annex B)

sk: Karakteristik zemin kar yükü değeri, is the characteristic value of snow load on the ground

Ce: Maruz kalma katsayısı, is the exposure coefficient

Ct: Isı katsayısı is the thermal coefficient

Rüzgar Yükü /Wind Load



Rüzgar Yükü /Wind Load

TS EN 1991-1-4 Madde 5.3 /TS EN 1991-1-4 Item 5.3

- $Fw = cscd \cdot cf \cdot qp\,(ze\,) \cdot Aref$

Burada;

- cscd: Kısım 6'da tanımlanan yapısal katsayı, is the structural factor as defined in Section 6
- cf: Kısım 7 veya Kısım 8'de verilen, yapı veya yapısal eleman için kuvvet katsayısı,

  is the force coefficient for the structure or structural element, given in Section 7 or Section 8

- qp(ze): ze referans yüksekliğindeki (Kısım 7 veya Kısım 8'de tanımlanmıştır) tepe rüzgâr basıncı (Madde 4.5'te tanımlanmıştır),

  is the peak velocity pressure (defined in 4.5) at reference height z_e (defined in Section 7 or Section 8)

- Aref: Yapının veya yapısal elemanın referans alanı (Kısım 7 veya Kısım 8'de verilmiştir)

dır. is the reference area of the structure or structural element, given in Section 7 or Section 8

Yük kombinasyonları /Load Combinations

G : sabit yük *(Dead Load)*
Q : hareketli yük *(Live Load)*
Qr : çatı hareketli yükü *(Roof Live Load)*
S : kar yükü *(Snow Load)*
R : yağmur yükü *(Rain Load)*
W : rüzgar yükü *(Wind Load)*
E : deprem etkisi *(Earthquake Load)*
F : akışkan madde basınç yükü *(Liquid Pressure Load)*
T : sıcaklık değişmesi veya mesnet çökmesi etkileri *(Thermal Load or Effect of Deflection of Bearing Support)*
H : yatay zemin basıncı, zemin suyu basıncı veya yığılı madde basıncı *(Lateral Soil Pressure Load, Ground Water Pressure Load, or Stacked Material Pressure Load)*

5.3.1 – Yük ve Dayanım Katsayıları ile Tasarım (YDKT) ***/Load and Resistance Factor Design (LRFD)***

Bu tasarım yönteminde gerekli dayanım, *R*u, aşağıdaki yük birleşimleri ile belirlenecektir. *(In this design method, the required strength, Ru, will be determined by the following load combinations.)*

(1) 1.4*G*
(2) 1.2*G* + 1.6*Q* + 0.5(Qr veya *S* veya *R*)
(3) 1.2*G* + 1.6(Qr veya *S* veya *R*) + (*Q* veya 0.8*W*)
(4) 1.2*G* + 1.0*Q* + 0.5(Qr veya *S* veya *R*) + 1.6*W*
(5) 1.2*G* + 1.0*Q* + 0.2*S* + 1.0*E*
(6) 0.9*G* + 1.6*W*
(7) 0.9*G* + 1.0*E*

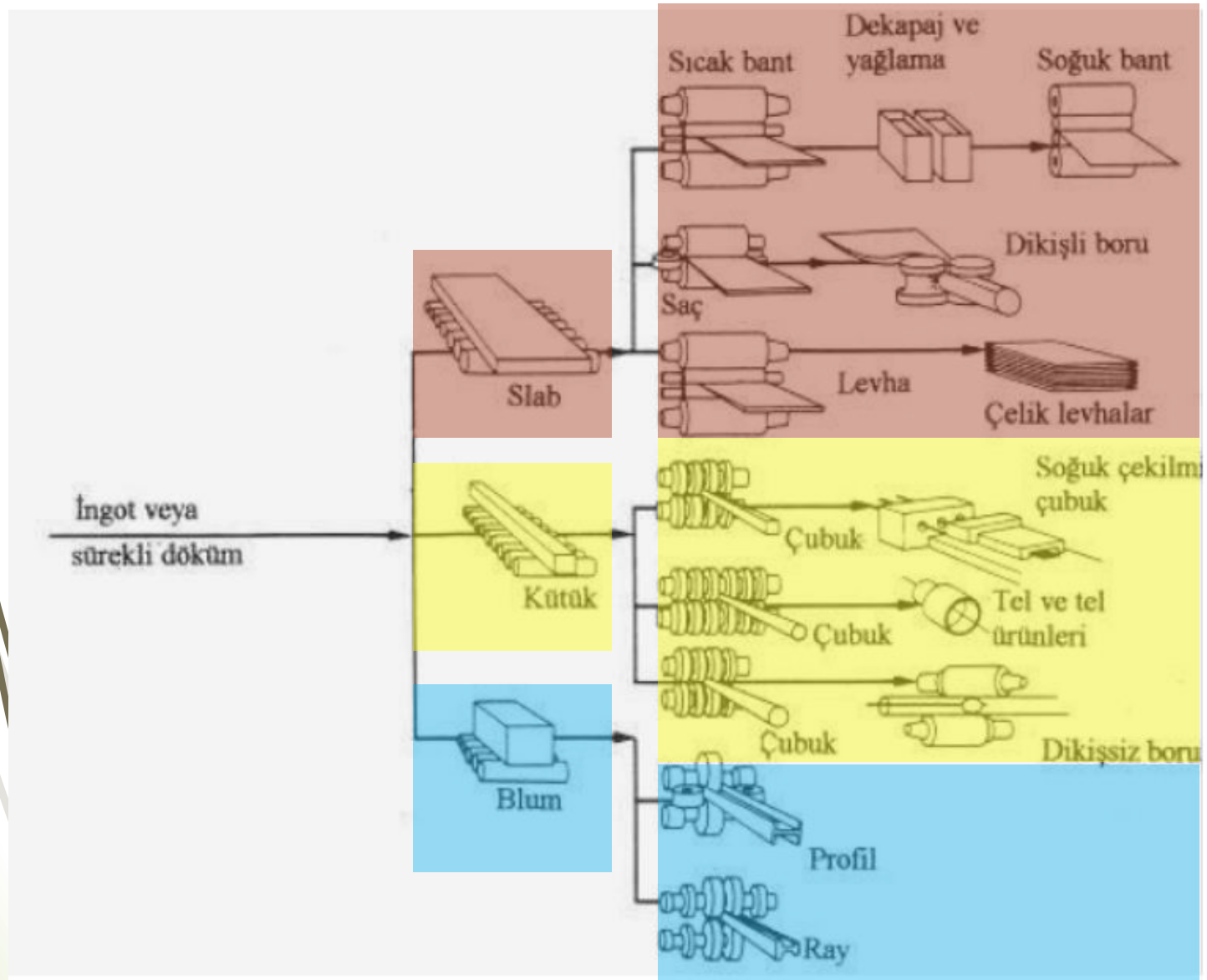
Not:

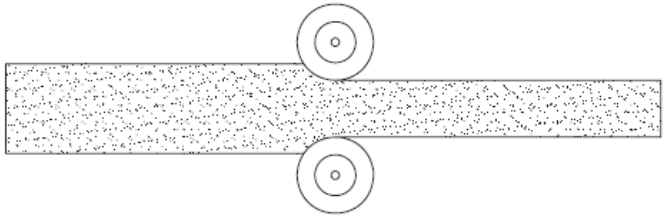
(a) *F* akışkan madde basınç yükünün mevcut olması halinde, (1) – (4) sayılı birleşimlerde bu yük *G* yükünün katsayısı ile birleşime girecektir. *(If the fluid pressure F is present, in the combinations (1) - (4), this load will combine with the coefficient of the load G.)*

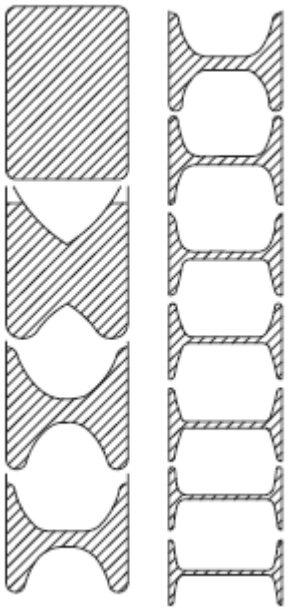
(b) *H* yatay kuvvetinin mevcut olması halinde, bu etki gerekli dayanımı arttıracak yönde ise 1.6 katsayısı ile, gerekli dayanımı azaltacak yönde olması halinde ise 0.9 katsayısı ile birleşime girecektir. *H* yatay kuvvetinin deprem etkisini içeren yük birleşimlerine katkısı, DBYBHY kapsamında değerlendirilecektir. *(If the horizontal force H is present, this effect will combine with a coefficient of 1.6 if it is in the direction to increase the required strength, and with a coefficient of 0.9 if it is in the direction of decreasing the required strength. The contribution of the horizontal force H to load combinations including earthquake effect will be evaluated within the scope of DBYBHY.)*

(c) *T* sıcaklık değişmesi ve/veya mesnet çökmesi etkilerinin mevcut olması halinde, bu etkiler gerekli dayanımı arttıracak yönde ise 1.0 katsayısı ile tüm birleşimlere girecektir. *(In the presence of T temperature change and / or effects of deflection of bearing support, these effects will enter all joints with a coefficient of 1.0 in the direction of increasing the required strength.)*

Malzeme hazırlanması /Production Process







Malzeme hazırlanması /Production Process

Youtube Videos

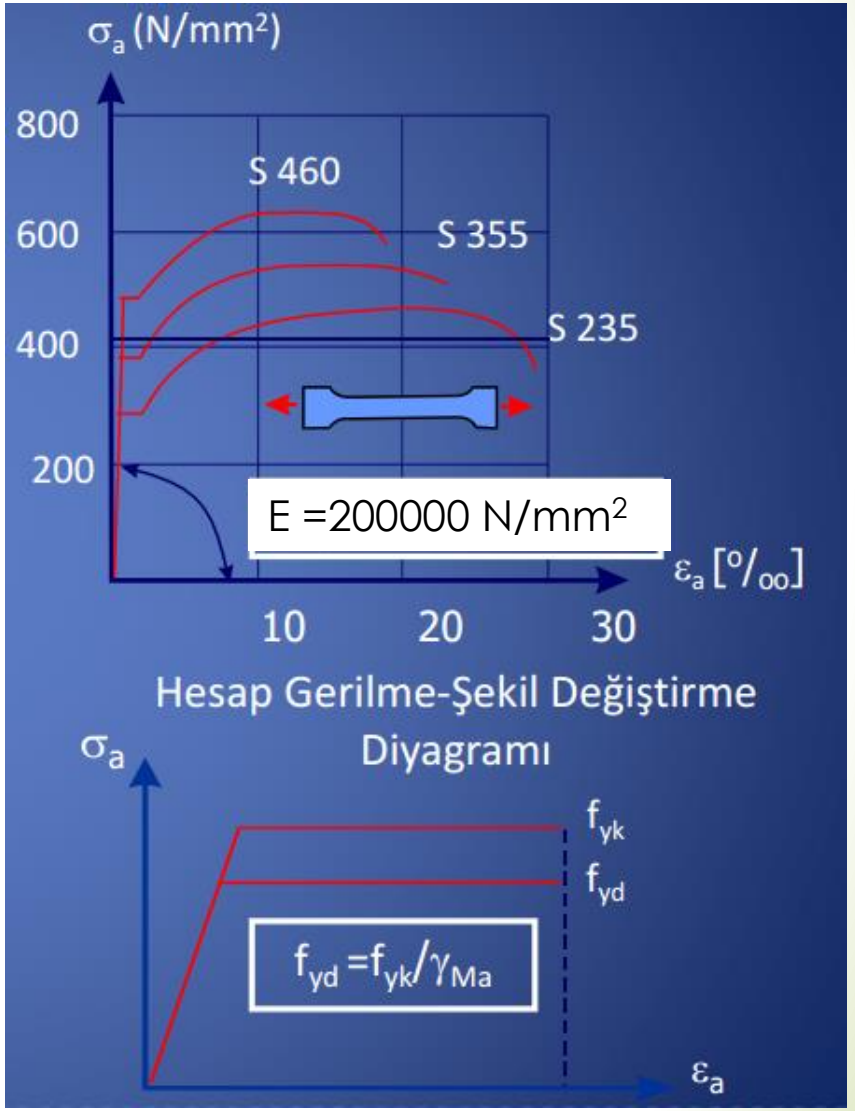
- Steelmaking: Rolling, https://www.youtube.com/watch?v=6xnKmt_gsLs
- STEEL: From Start to Finish, https://www.youtube.com/watch?v=9I7JqonyoKA
- Steelmaking, https://www.youtube.com/watch?v=Ea_7Rnd8BTM
- Hot Rolling, Cold Rolling & Grain Flow in Rolling Process, https://www.youtube.com/watch?v=JIM_t9QH8J8
- Seamless steel tubes production process, https://www.youtube.com/watch?v=Rhc_Kkc-vMY
- Integrated tube production at Sandvik, https://www.youtube.com/watch?v=7ITgMXtuMv0
- Amazing Biggest Steel Bending Machine At Work, Fast Extreme Large Plate Rolling Machines, https://www.youtube.com/watch?v=qkuxjLKKbWA

Çelik Malzeme Özellikleri /Material Properties of Steel

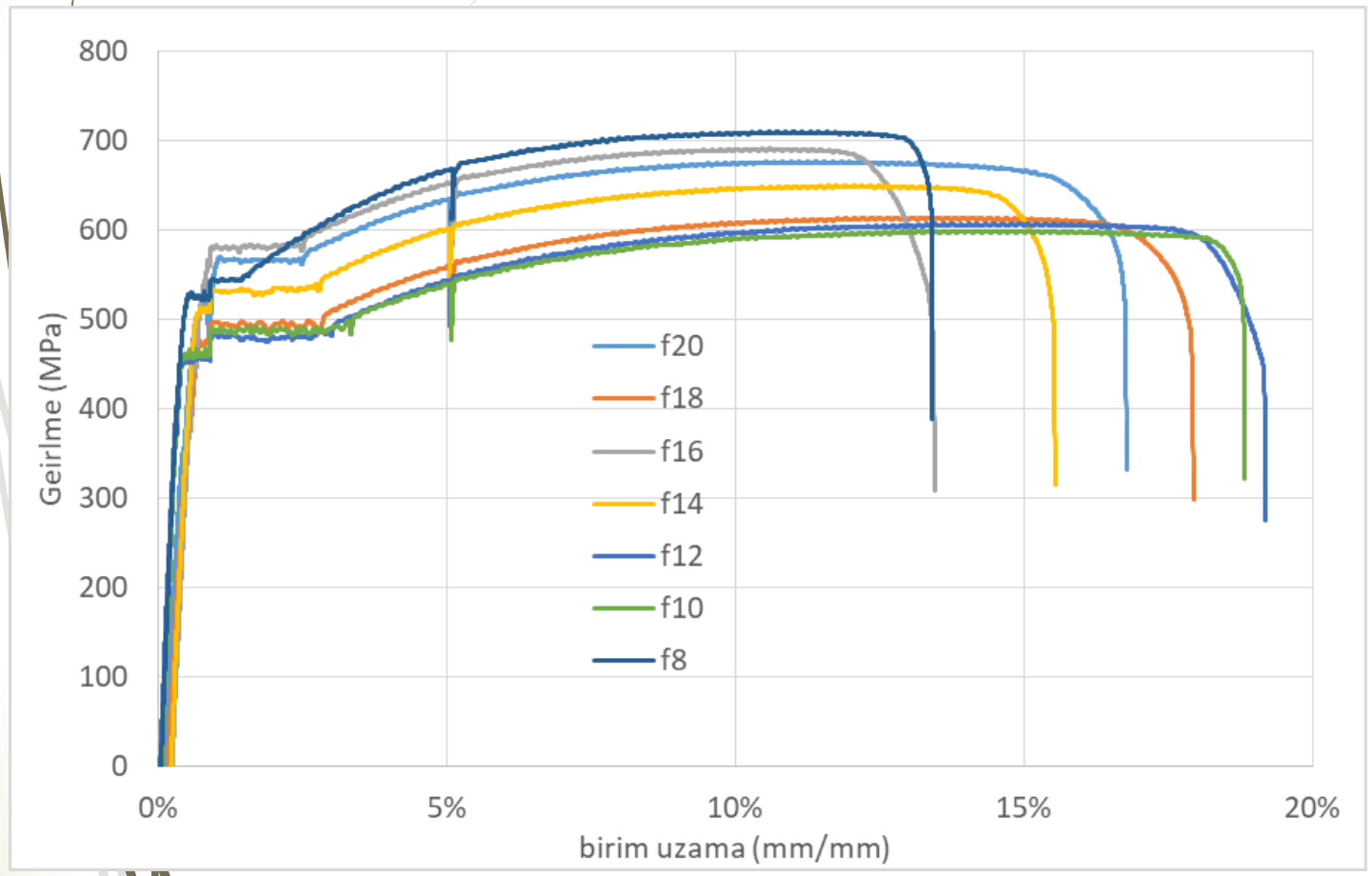
TABLO 2.1A – SICAK HADDELENMİŞ YAPISAL ÇELİKLERDE KARAKTERİSTİK AKMA GERİLMESİ, F_y VE ÇEKME DAYANIMI, F_u

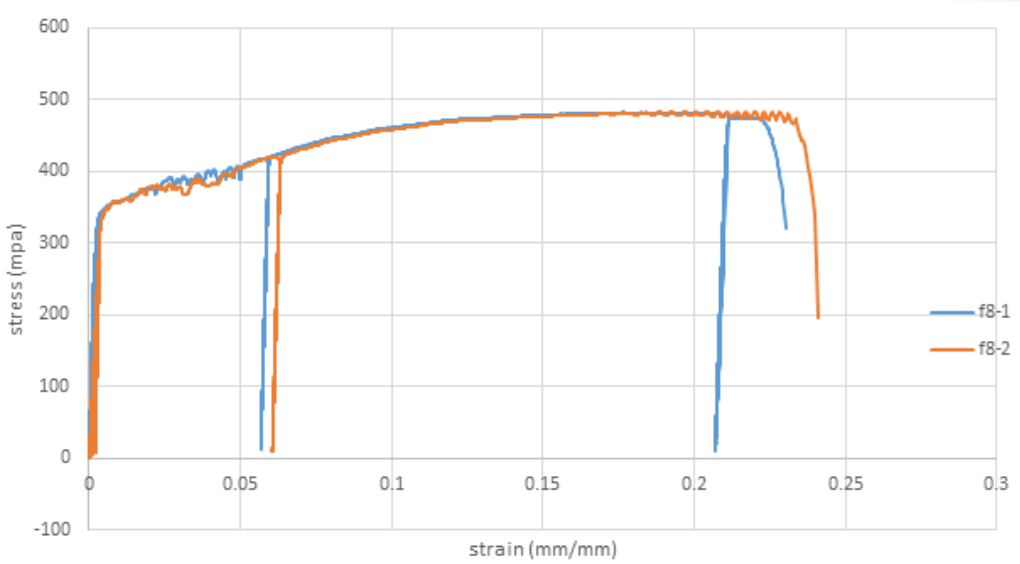
Standart ve Çelik Sınıfı	Karakteristik Kalınlık, t (mm)			
	$t \leq 40$mm		40mm $< t \leq$ 80mm	
	F_y (N/mm^2)	F_u (N/mm^2)	F_y (N/mm^2)	F_u (N/mm^2)
EN 10025-2				
S235	235	360	215	360
S275	275	430	255	410
S355	355	510	335	470
S450	440	550	410	550
EN 10025-3				
S275 N/NL	275	390	255	370
S355 N/NL	355	490	335	470
S420 N/NL	420	520	390	520
S460 N/NL	460	540	430	540
EN 10025-4				
S275 M/ML	275	370	255	360
S355 M/ML	355	470	335	450

Norm	EN 10025	DIN 17100	TS 648
Gösterim	S235	St 37	Fe 37
	S355	St 52	Fe 52



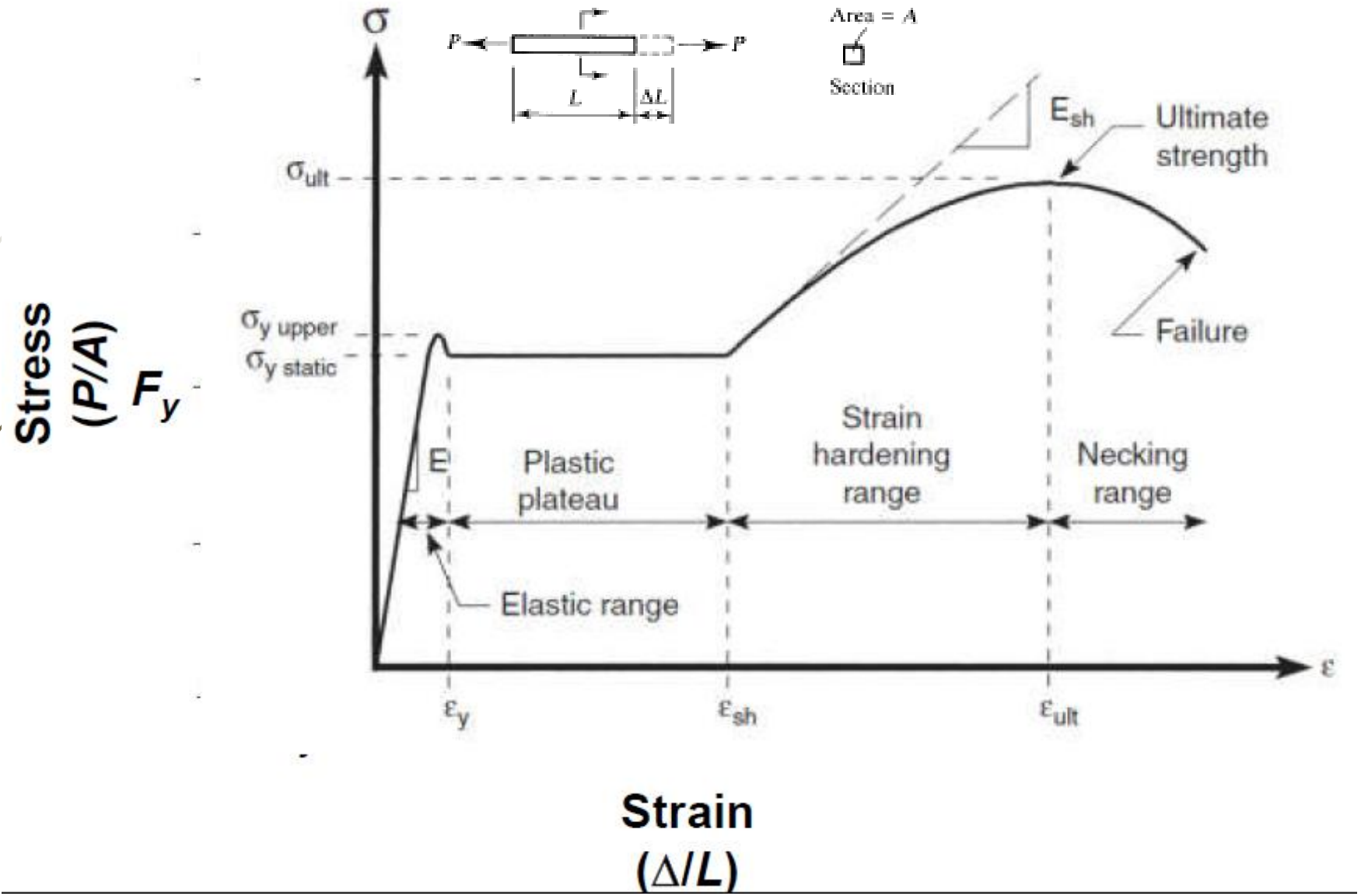
Çelik Malzeme Özellikleri /Material Properties of Steel







Çelik Malzeme Özellikleri /Material Properties of Steel



5.2.2 – Yük ve Dayanım Katsayıları ile Tasarım (YDKT) ***(Load and Resistance Factor Design (LRFD))***
Yük ve Dayanım Katsayıları ile Tasarım (YDKT), tüm yapısal elemanlar için, *tasarım dayanımı*, ϕR_n nin bu tasarım yöntemi için öngörülen ve **Bölüm 5.3.1** de verilen YDKT yük birleşimleri altında hesaplanan *gerekli dayanım*, R_u, değerine eşit veya daha büyük olması prensibine dayanmaktadır. *(Load and Resistance Factor Design (LRFD) is based on the principle that the design strength, ϕR_n , for all structural members is equal to or greater than the required strength, Ru, calculated under the LRFD load combinations for this design method and given in Section 5.3.1.)*

Buna göre, tasarım **Denk.(5.1)** de verilen koşula uygun olarak gerçekleştirilecektir. *(Therefore, the design will be carried out in accordance with the condition given in Equation (5.1).)*

$$\mathbf{R_u \leq \phi R_n}$$

Burada,

R_u : YDKT yük birleşimleri altında hesaplanan gerekli dayanım. *(Required strength calculated under LRFD load combinations.)*
R_n : Karakteristik dayanım. *(Nominal strength)*
ϕ : Dayanım katsayısı. *(Resistance factor)*
ϕR_n : Tasarım dayanımı. *(Design strength)*

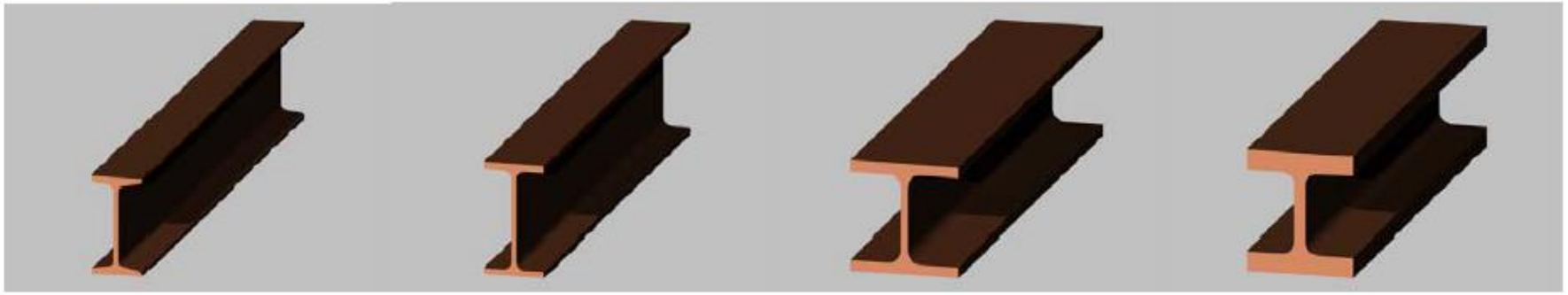
Hadde profiller /Hot-rolled Profiles



Profil		U 200	I 200	IPEa 200	IPE 200	IPEo 200	HEAA-200	HEA-200	HEB-200	HEM-200
h	mm	200	200	197	200	202	186	190	200	220
b	mm	75	90	100	100	102	200	200	200	206
s	mm	8,5	7,5	4,5	5,6	6,2	5,5	6,5	9	15
t	mm	11,5	11,3	7,2	8,5	9,5	8	10	15	25
r	mm	6	4,5	12	12	12	18	18	18	18
A	cm^2	32,2	33,4	23,5	28,5	32,0	44,1	53,8	78,1	131
g	kg/m	25,3	26,2	18,4	22,4	25,1	34,6	42,3	61,3	103
I_y	cm^4	1910	2140	1592	1940	2210	2944	3690	5700	10640
W_y	cm^3	191	214	162	194	219	317	389	570	967
I_z	cm^4	148	117	117	142	169	1068	1340	2000	3650
W_z	cm^3	27,0	26,0	23,4	28,5	33,1	107	134	200	354
I_T	cm^4	11,9	13,5	4,11	7,02	9,41	12,7	21,1	59,5	260
C_M	cm^6	9070	10520	10500	12990	15570	84500	108000	171100	346300
$N_{pl,k}$	kN	773	802	563	684	767	1059	1292	1874	3151
$V_{pl,z,k}$	kN	222	196	118	149	165	136	162	231	405
$M_{pl,y,k}$	kNm	57,2	59,7	43,6	53,0	59,9	83,3	103	154	272
$M_{pl,y}/M_{el,y}$		1,247	1,162	1,121	1,138	1,140	1,095	1,103	1,126	1,172

Örnek Hadde Profiller /Examples of Hot-rolled Profiles

- **I Profilleri**

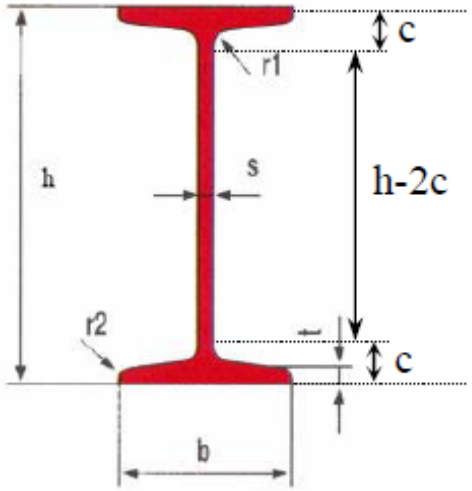


Normal I
(I veya IPN)

Orta Genişlikte I (IPE)

Geniş Başlıklı I
(IPB veya HEB)

Geniş Başlıklı Ağır I
(IPBv veya HEM)



Örnek: I 300

h=300 mm

b=125 mm

$s=r_1=10,8$ mm

t=16,2 mm

$r_2=6,5$ mm

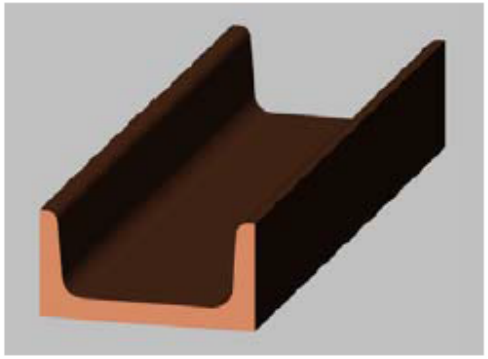
c= 29,5 mm

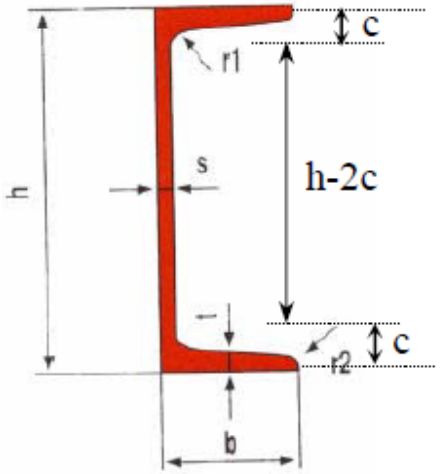
h-2c=241 mm

Profil boyut aralıkları:
I 80 – I 600
IPE 80- IPE 600
IPB 100- IPB 1000
IPBv 100- IPBv 1000

Örnek Hadde Profiller /Examples of Hot-rolled Profiles

- **[Profilleri**



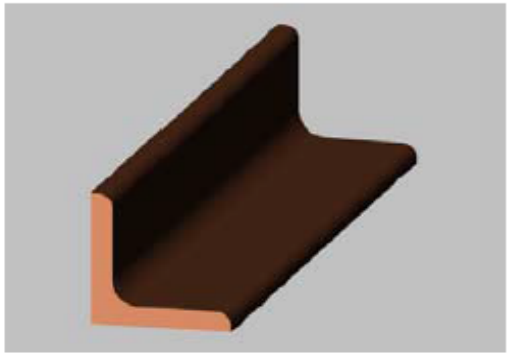


Örnek: U 200
h=200 mm
b=75 mm
s=8,5 mm
$t=r_1=11,5$ mm
$r_2=6$ mm
c=24,5 mm
h-2c=151 mm

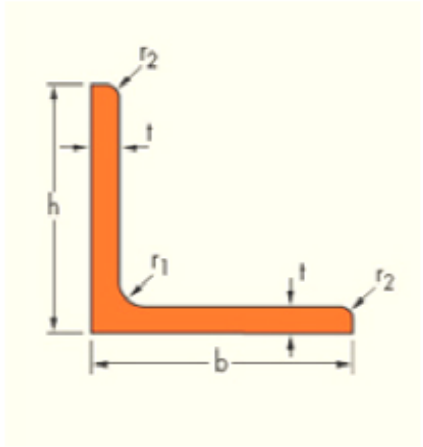
Profil boyut aralıkları:
U 30 – U 400

Örnek Hadde Profiller /Examples of Hot-rolled Profiles

- **L Profilleri (Korniyerler)**

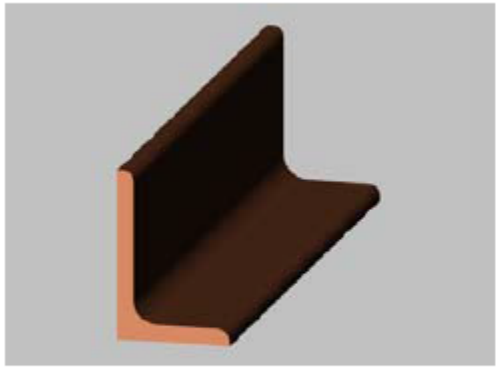


Eşit kollu korniyer

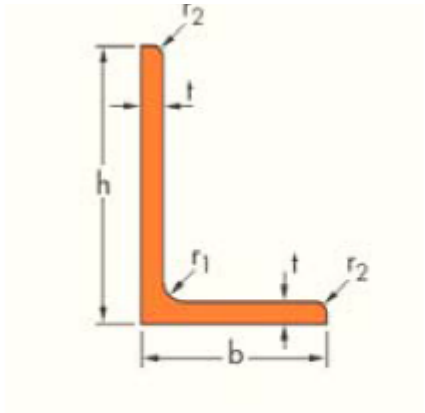


Örnek: L 50x5
h=b=50 mm
t=5 mm
r_1=7 mm
r_2=3,5 mm

Profil boyut aralıkları:
L 20x3 – L 250x24



Farklı kollu korniyer



Örnek: L 150x75x9
h=150 mm
b=75 mm
t=9 mm
r_1=10,5 mm
r_2=5,5 mm

Profil boyut aralıkları:
L 30x20x3 – L 250x90x16

Sözlük /Glossary

- **Narinlik**; uzunluk kalınlık oranı veya uzunluk atalet yarıçapı oranı
- **Atalet Yarıçapı**; Atalet Momentinin alana oranının karekökü
- **Burkulma**; eleman veya eleman parçasının basınç yükü altında dayanımı kaybetmesi
- **Burkulma katsayısı**; elemanın mesnet koşullarının etkili boydaki oranı
- **Bölgesel burkulma**; elemanın çok küçük bir bölgesinde meydana gelen burkulma
- **Kompak**; narinlik değerinin burkulma oluşturmayacak dercede olması
- **Narin**; narinlik değerinin büyük olması
- **Kampak olmayan**; elemanın narinlik değerinin kompak ile narin arasında olması
- **Plastik mafsal**; elemanın bir noktasında dönme değerinin diğer noktalara göre çok fazla olması

Sözlük /Glossary

*Allowable strength**†. Nominal strength divided by the safety factor, R_n/Ω.

*Allowable stress**. Allowable strength divided by the applicable section property, such as section modulus or cross-sectional area.

ASD (allowable strength design)†. Method of proportioning structural components such that the allowable strength equals or exceeds the required strength of the component under the action of the ASD load combinations.

*Available strength**†. Design strength or allowable strength, as applicable.

*Available stress**. Design stress or allowable stress, as applicable.

Bearing-type connection. Bolted connection where shear forces are transmitted by the bolt bearing against the connection elements.

Block shear rupture†. In a connection, limit state of tension rupture along one path and shear yielding or shear rupture along another path.

Braced frame†. Essentially vertical truss system that provides resistance to lateral forces and provides stability for the structural system.

Buckling†. Limit state of sudden change in the geometry of a structure or any of its elements under a critical loading condition.

Buckling strength. Strength for instability limit states.

Built-up member, cross section, section, shape. Member, cross section, section or shape fabricated from structural steel elements that are welded or bolted together.

Sözlük /Glossary

Cold-formed steel structural member†. Shape manufactured by press-braking blanks sheared from sheets, cut lengths of coils or plates, or by roll forming cold- or hot-rolled coils or sheets; both forming operations being performed at ambient room temperature, that is, without manifest addition of heat such as would be required for hot forming.

Collector. Also known as drag strut; member that serves to transfer loads between floor diaphragms and the members of the lateral force-resisting system.

Compact section. Section capable of developing a fully plastic stress distribution and possessing a rotation capacity of approximately three before the onset of local buckling.

Cover plate. Plate welded or bolted to the flange of a member to increase cross-sectional area, section modulus or moment of inertia.

*Design strength**†. Resistance factor multiplied by the nominal strength, ϕR_n.

Effective length factor, K. Ratio between the effective length and the unbraced length of the member.

Effective length. Length of an otherwise identical compression member with the same strength when analyzed with simple end conditions.

Effective net area. Net area modified to account for the effect of shear lag.

Effective section modulus. Section modulus reduced to account for buckling of slender compression elements.

Effective width. Reduced width of a plate or slab with an assumed uniform stress distribution which produces the same effect on the behavior of a structural member as the actual plate or slab width with its nonuniform stress distribution.

Sözlük /Glossary

Fastener. Generic term for bolts, rivets or other connecting devices.

Filler metal. Metal or alloy added in making a welded joint.

Filler. Plate used to build up the thickness of one component.

Fillet weld reinforcement. Fillet welds added to groove welds.

Fillet weld. Weld of generally triangular cross section made between intersecting surfaces of elements.

Flexural buckling†. Buckling mode in which a compression member deflects laterally without twist or change in cross-sectional shape.

Flexural-torsional buckling†. Buckling mode in which a compression member bends and twists simultaneously without change in cross-sectional shape.

Groove weld. Weld in a groove between connection elements. See also AWS D1.1/D1.1M.

Gusset plate. Plate element connecting truss members or a strut or brace to a beam or column.

Lateral-torsional buckling†. Buckling mode of a flexural member involving deflection out of the plane of bending occurring simultaneously with twist about the shear center of the cross section.

*Local buckling***. Limit state of buckling of a compression element within a cross section.

*Local yielding***†. Yielding that occurs in a local area of an element.

LRFD (load and resistance factor design)†. Method of proportioning structural components such that the design strength equals or exceeds the required strength of the component under the action of the LRFD load combinations.

Sözlük /Glossary

*Nominal strength**†. Strength of a structure or component (without the resistance factor or safety factor applied) to resist load effects, as determined in accordance with this Specification.

Noncompact section. Section that is able to develop the yield stress in its compression elements before local buckling occurs, but is unable to develop a rotation capacity of three.

Nondestructive testing. Inspection procedure wherein no material is destroyed and the integrity of the material or component is not affected.

Plastic hinge. Fully yielded zone that forms in a structural member when the plastic moment is attained.

Plastic moment. Theoretical resisting moment developed within a fully yielded cross section.

Prying action. Amplification of the tension force in a bolt caused by leverage between the point of applied load, the bolt, and the reaction of the connected elements.

Punching load. In an HSS connection, component of branch member force perpendicular to a chord.

P-δ effect. Effect of loads acting on the deflected shape of a member between joints or nodes.

P-Δ effect. Effect of loads acting on the displaced location of joints or nodes in a structure. In tiered building structures, this is the effect of loads acting on the laterally displaced location of floors and roofs.

Quality assurance. Monitoring and inspection tasks to ensure that the material provided and work performed by the fabricator and erector meet the requirements of the approved construction documents and referenced standards. Quality assurance includes those tasks designated "special inspection" by the applicable building code.

Sözlük /Glossary

*Required strength**†. Forces, stresses and deformations acting on a structural component, determined by either structural analysis, for the LRFD or ASD load combinations, as applicable, or as specified by this Specification or Standard.

Resistance factor, ϕ†. Factor that accounts for unavoidable deviations of the nominal strength from the actual strength and for the manner and consequences of failure.

Rupture strength†. Strength limited by breaking or tearing of members or connecting elements.

Second-order effect. Effect of loads acting on the deformed configuration of a structure; includes P-δ effect and P-Δ effect.

Seismic force-resisting system. That part of the structural system that has been considered in the design to provide the required resistance to the seismic forces prescribed in ASCE/SEI 7.

Shear lag. Nonuniform tensile stress distribution in a member or connecting element in the vicinity of a connection.

Shear wall†. Wall that provides resistance to lateral loads in the plane of the wall and provides stability for the structural system.

Shear yielding (punching). In an HSS connection, limit state based on out-of-plane shear strength of the chord wall to which branch members are attached.

Sidesway buckling (frame). Stability limit state involving lateral sidesway instability of a frame.

Simple connection. Connection that transmits negligible bending moment between connected members.

Sözlük /Glossary

Slender-element section. Cross section possessing plate components of sufficient slenderness such that local buckling in the elastic range will occur.

Slip. In a bolted connection, limit state of relative motion of connected parts prior to the attainment of the available strength of the connection.

Slip-critical connection. Bolted connection designed to resist movement by friction on the faying surface of the connection under the clamping force of the bolts.

Slot weld. Weld made in an elongated hole fusing an element to another element.

Snug-tightened joint. Joint with the connected plies in firm contact as specified in Chapter J.

Specifications. Written documents containing the requirements for materials, standards and workmanship.

Specified minimum tensile strength. Lower limit of tensile strength specified for a material as defined by ASTM.

Specified minimum yield stress†. Lower limit of yield stress specified for a material as defined by ASTM.

Splice. Connection between two structural elements joined at their ends to form a single, longer element.

Stability. Condition in the loading of a structural component, frame or structure in which a slight disturbance in the loads or geometry does not produce large displacements.

Steel anchor. Headed stud or hot rolled channel welded to a steel member and embedded in concrete of a composite member to transmit shear, tension, or a combination of shear and tension at the interface of the two materials.

Stiffened element. Flat compression element with adjoining out-of-plane elements along both edges parallel to the direction of loading.

Stiffener. Structural element, typically an angle or plate, attached to a member to distribute load, transfer shear or prevent buckling.

Sözlük /Glossary

Stiffness. Resistance to deformation of a member or structure, measured by the ratio of the applied force (or moment) to the corresponding displacement (or rotation).

Story drift. Horizontal deflection at the top of the story relative to the bottom of the story.

Story drift ratio. Story drift divided by the story height.

Stress concentration. Localized stress considerably higher than average due to abrupt changes in geometry or localized loading.

Strong axis. Major principal centroidal axis of a cross section.

Tensile strength (of material)†. Maximum tensile stress that a material is capable of sustaining as defined by ASTM.

Tensile strength (of member). Maximum tension force that a member is capable of sustaining.

Tension and shear rupture†. In a bolt or other type of mechanical fastener, limit state of rupture due to simultaneous tension and shear force.

Torsional buckling†. Buckling mode in which a compression member twists about its shear center axis.

Transverse reinforcement. In an encased composite column, steel reinforcement in the form of closed ties or welded wire fabric providing confinement for the concrete surrounding the steel shape.

Transverse stiffener. Web stiffener oriented perpendicular to the flanges, attached to the web.

Turn-of-nut method. Procedure whereby the specified pretension in high-strength bolts is controlled by rotating the fastener component a predetermined amount after the bolt has been snug tightened.

Unbraced length. Distance between braced points of a member, measured between the centers of gravity of the bracing members.

Sözlük /Glossary

Unstiffened element. Flat compression element with an adjoining out-of-plane element along one edge parallel to the direction of loading.

Weak axis. Minor principal centroidal axis of a cross section.

Yield moment†. In a member subjected to bending, the moment at which the extreme outer fiber first attains the yield stress.

Yield point†. First stress in a material at which an increase in strain occurs without an increase in stress as defined by ASTM.

Yield strength†. Stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain as defined by ASTM.

Yield stress†. Generic term to denote either yield point or yield strength, as applicable for the material.

Yielding†. Limit state of inelastic deformation that occurs when the yield stress is reached.

Yielding (plastic moment)†. Yielding throughout the cross section of a member as the bending moment reaches the plastic moment.

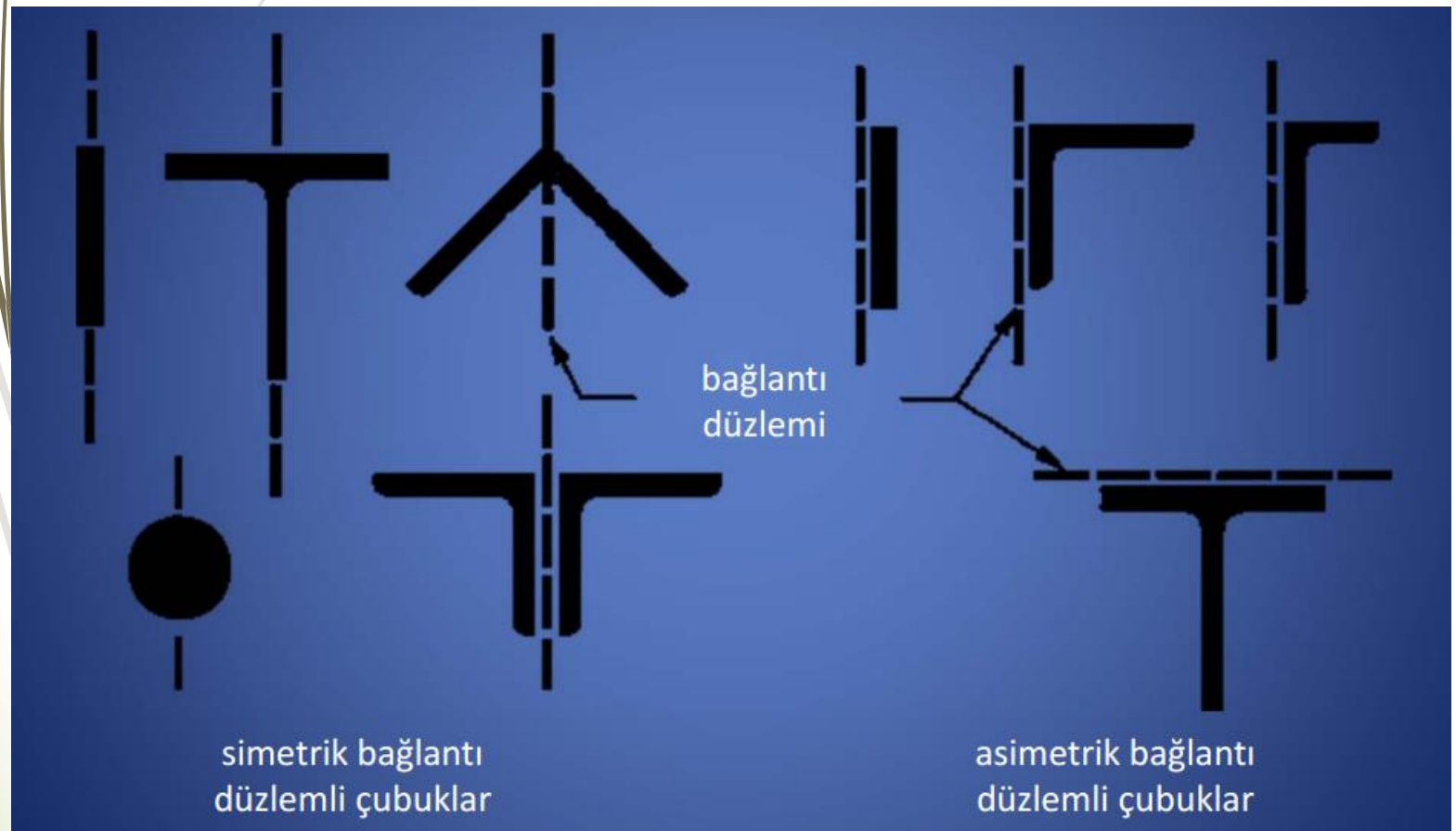
Yielding (yield moment)†. Yielding at the extreme fiber on the cross section of a member when the bending moment reaches the yield moment.

Kısaltmalar /Abbreviations

ACI (American Concrete Institute)
AISC (American Institute of Steel Construction)
ANSI (American National Standards Institute)
ASCE (American Society of Civil Engineers)
ASD (allowable strength design)
ASME (American Society of Mechanical Engineers)
ASNT (American Society for Nondestructive Testing)
AWI (associate welding inspector)
AWS (American Welding Society)
CJP (complete joint penetration)
CVN (Charpy V-notch)
ERW (electric resistance welded)
FCAW (flux cored arc welding)
FR (fully restrained)
GMAW (gas metal arc welding)
HSS (hollow structural section)
LRFD (load and resistance factor design)
NDT (nondestructive testing)
PR (partially restrained)
QA (quality assurance)
QC (quality control)
QCI (quality control inspector)
RT (radiographic testing)
SAW (submerged arc welding)
SMAW (shielded metal arc welding)
UT (ultrasonic testing)
WI (welding inspector)

Bölüm 7. Eksenel Çekme Kuvveti Etkisi /Design of Members For Tension

Eksenel çekme kuvveti etkisindeki elemanların tasarımı bu bölümde belirtilen kurallara göre yapılacaktır. *(The design of the elements under tension force will be made according to the rules specified in this section.)*





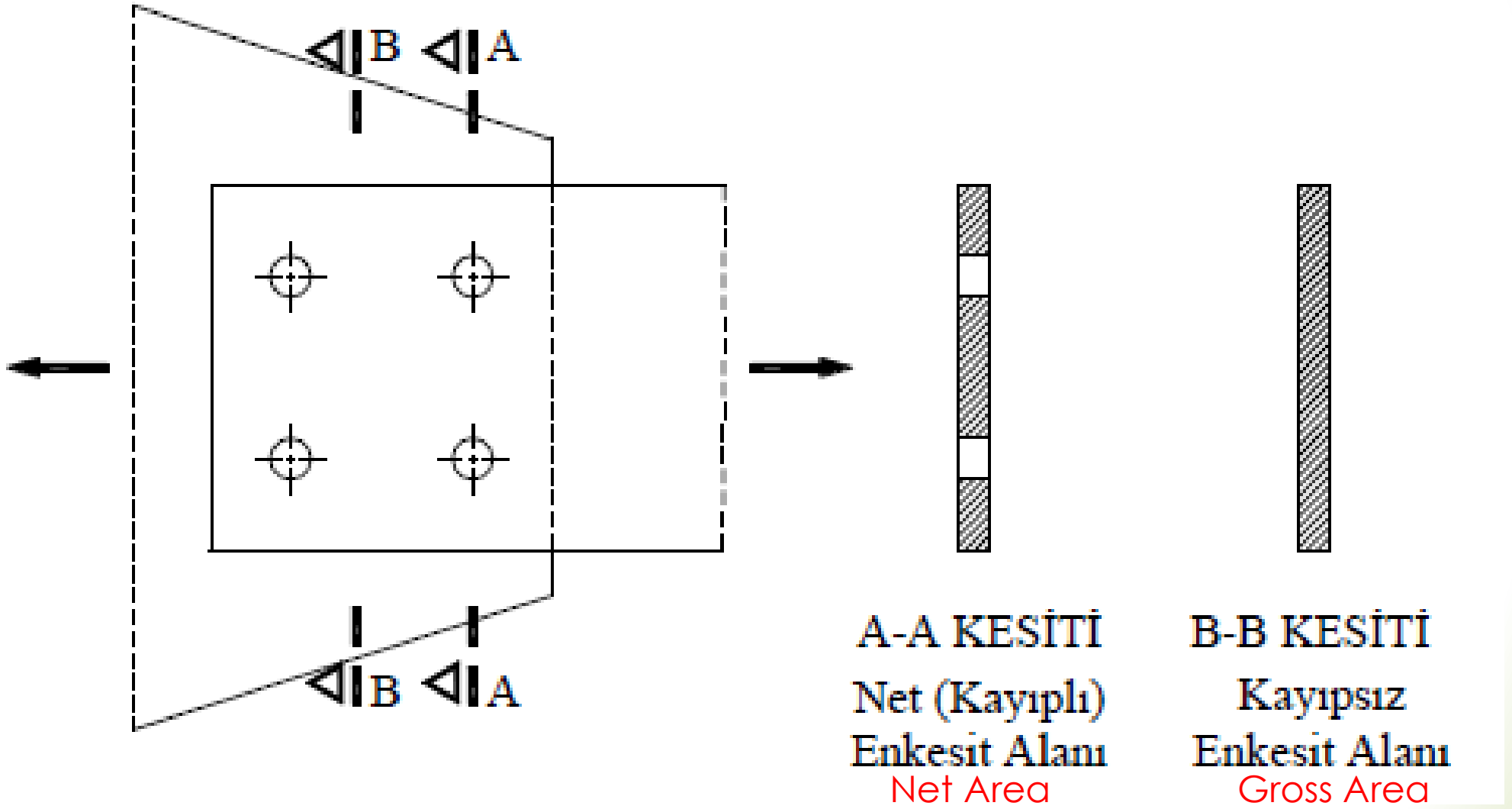


Çekme Elemanları /Tension Member

Çekme elemanları 4 (dört) ana guruba ayrılabilir: *(Tension Members can be orginized in 4 (four) main groups)*

- Teller, ipler, halatlar ve kablolar *(wire, strands, rope, and cables)*
- Çubuklar *(rods or bars)*
- Tek parçalı çekme elemanları *(Single tension members)*
- Çok parçalı çekme elemanları *(Latticed tension members)*

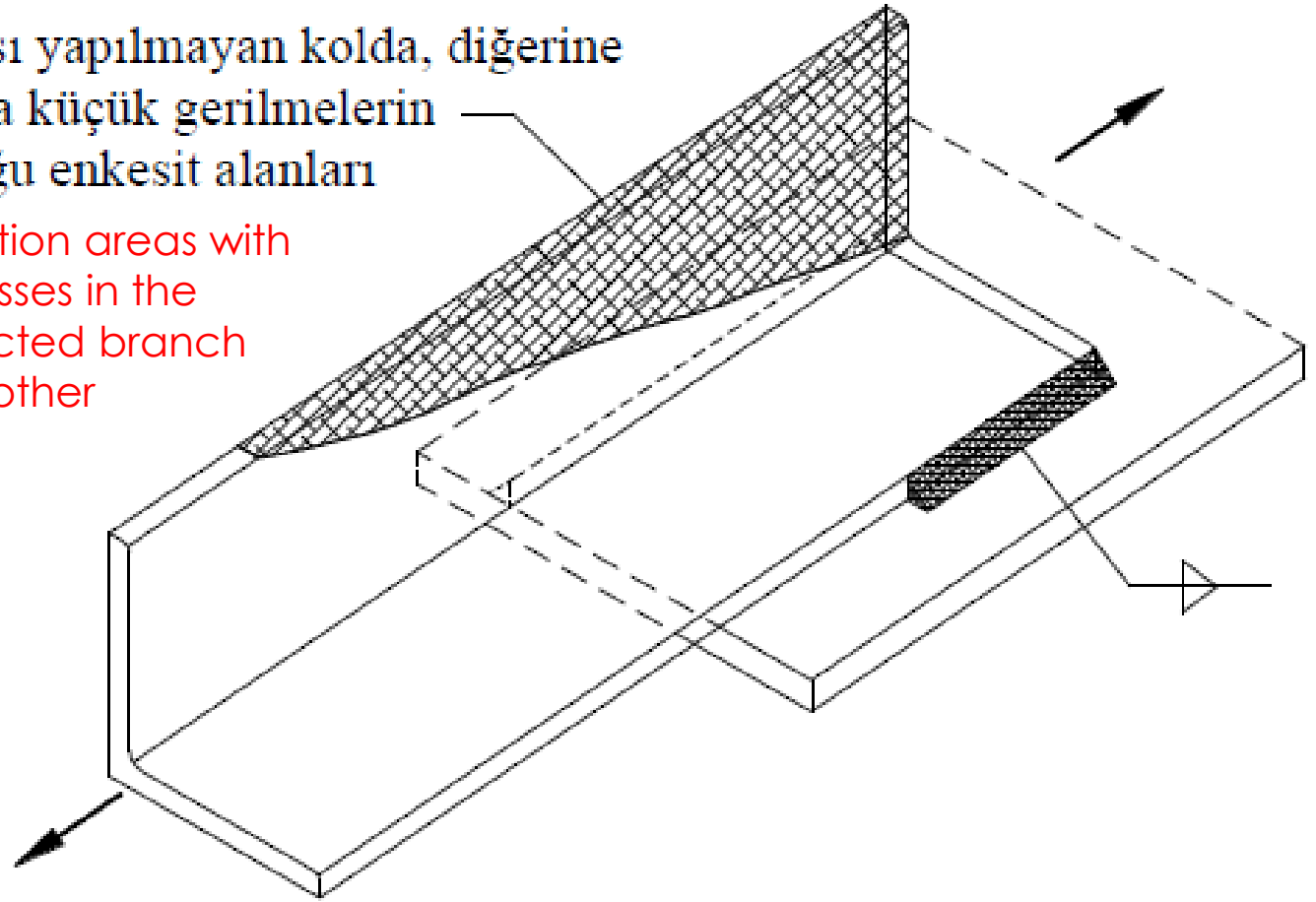
Toplam alan ve net alan /Gross Area and Net Area



Gerilme Yığışımı /Stress Consentration

Bağlantısı yapılmayan kolda, diğerine göre daha küçük gerilmelerin bulunduğu enkesit alanları

Cross-section areas with lower stresses in the unconnected branch than the other



Etkin Net Enkesit Alanı /Effective Net Area

- Çekme elemanının birleşim bölgesine tüm enkesit parçalarıyla bulonlu veya kaynaklı olarak bağlanmadığı durumda, gerilme yayılışındaki düzensizliğin hesaba katıldığı *etkin net enkesit alanı*, Ae, *Yönetmelik* 7.1.3 esas alınarak hesaplanacaktır. *(In case the tensile element is not bolted or welded to the junction with all cross-sectional parts, the effective net cross-sectional area , Ae where the irregularity in the stress propagation is taken into account shall be calculated on the basis of Regulation 7.1.3.)*

$$A_e = UA_n$$

- Korniyer enkesitli çekme elemanının sadece bir kolunun kaynaklı olarak bağlandığı Şekil 7.2 deki birleşim detayında, çekme elemanının bağlantısı yapılmayan koluna ait enkesitler daha küçük gerilme etkisinde kalmaları nedeniyle, birleşim bölgesinde ortaya çıkan *düzensiz gerilme dağılımı, gerilme düzensizliği etki katsayısı*, *U* ile gözönüne alınmaktadır. *(In the joint detail in Figure 7.2 where only one arm of the corner cross-section tensile element is connected by welding, the uneven stress distribution in the junction area is taken into consideration with shear lag factor as the cross sections belonging to the unconnected arm of the tensile element are under the effect of lower stress.)*

Çekme Elemanlarında Sınır Durumlar /Limit States In Tension Members

Üç farklı kapasite hesaplanır en küçüğü alınır

- Akma durumu *(Limit state of yielding)* için
  - Toplam alan dikkate alınarak akma dayanımı ile çarpılarak toplam kapasite bulunur
- Kırılma durumu *(Limit state of bearing)* için
  - Net efektif alan hesaplanıp maksimum(çekme) dayanım dikkate alınarak kapasite bulunur
- Blok kırılma *(Limit state of tension and shear rupture)* için
  - Farklı blok kırılma durumları için çekme ve kayma yüzeylerinde toplam kapasite bulunur

Basınç Çubukları /Compression Members

- Boyuna doğrultuda basınç kuvveti taşıyan çubuklara "basınç çubukları" denir. *(Bars or rods carrying pressure force in the longitudinal direction are called «compression member")*
- Kolonlar, basınç etkisindeki kafes kiriş elemanları basınç çubuklarına örnek olarak gösterilebilir. *(Columns are examples of compression member of truss elements under compression.)*
- İdeal şartlarda tam olarak ağırlık ekseninde tesir ettiği varsayılan eksenel kuvvet gerçekte var olamaz. Kesit zorunun çok küçük de olsa belli bir dış merkezlikle etki etmesi kaçınılmazdır. Bu dış merkezilik ise ufak da olsa bir eğilme etkisi yaratır, ancak ideal yükleme koşuluna yeterince yaklaşılmışsa bu etkiler ihmal edilir. *(Under ideal conditions, the axial force that is assumed to act exactly on the gravity axis cannot actually exist. It is inevitable that the axial force will affect with a certain eccentricity, even if the eccentricity is very small. The eccentricity creates a small bending effect, but if the ideal loading condition is sufficiently approached, the effect is neglected.)*

Basınç Çubukları /Compression Members

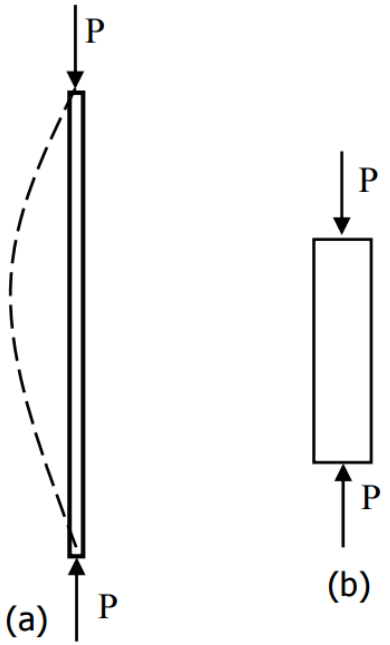
- Basınç çubuklarının taşıma gücünün hesabında burkulma olayı esas olduğundan bir "stabilite problemi" söz konusudur ve burkulma yükünün hesabı önem arz etmektedir. *(Since buckling is undeniable stuff in the calculation of the bearing strength of the compression members, there is a "stability problem" and the calculation of the buckling load is important.)*
- Merkezi basınç kuvveti etkisinde, malzemesi Hooke kanununa uyan, iki ucu mafsallı, prizmatik bir çubuk için ideal eğilme burkulması yükü ilk olarak 1757'de İsviçreli matematikçi Leonhard Euler tarafından hesaplanmıştır. *(The ideal bending buckling load for a prismatic rod with articulated two ends under the effect of central pressure force was first calculated by the Swiss mathematician Leonhard Euler in 1757.)*

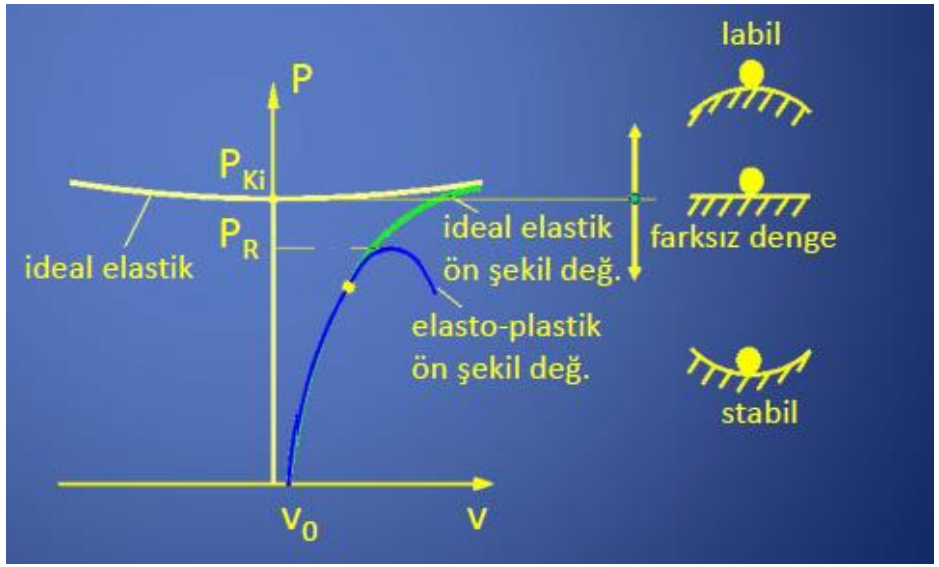
Three-Dimensional Model Viewer for Illustrating Structural Steel Concepts

- https://aisc-org.github.io/aisc-model-viewer/

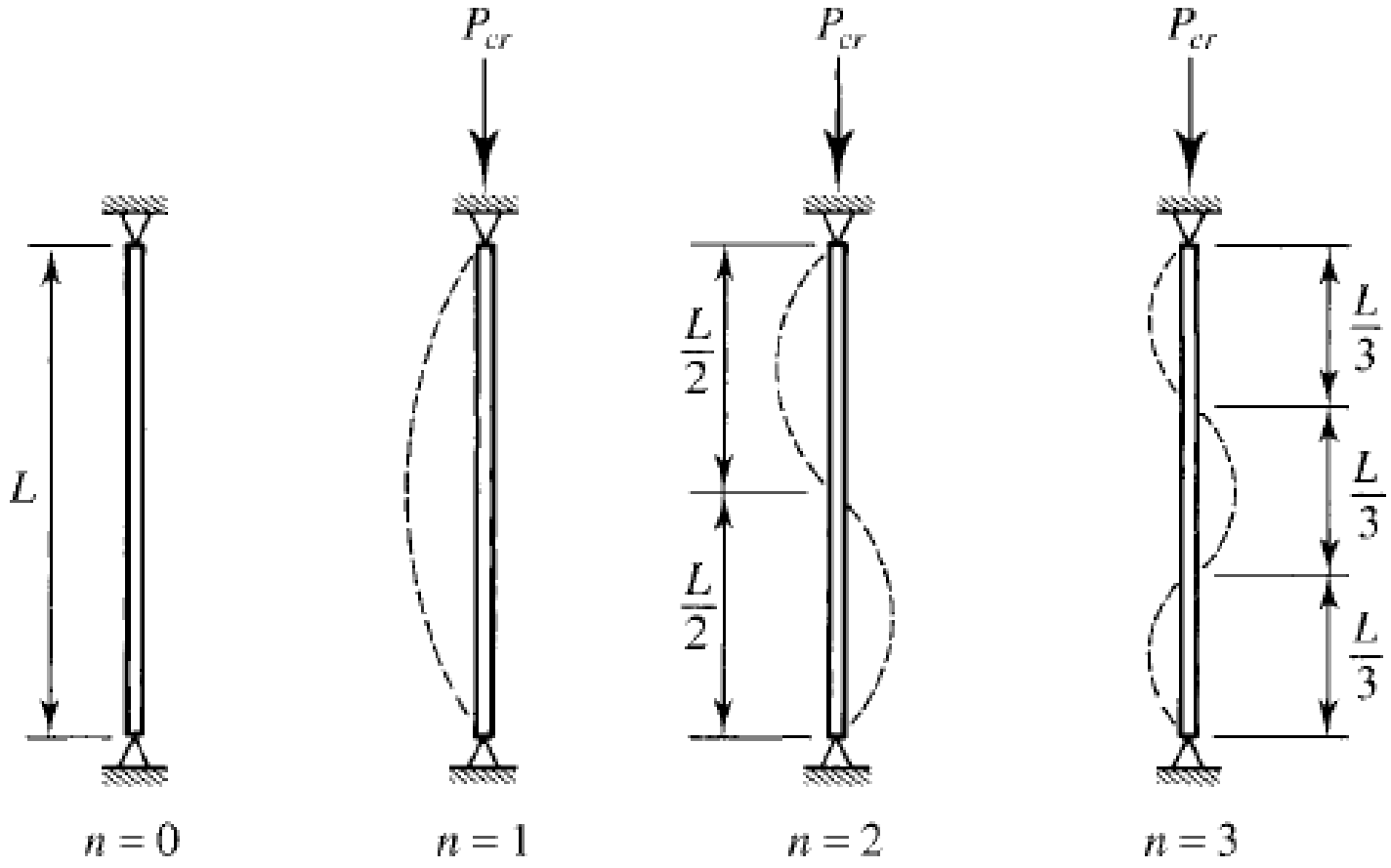
Basınç Çubukları /Compression Members

Euler burkulma yükü: $P_{cr,e} = \dfrac{\pi^2 EI}{L^2}$ L: burkulma boyu





Burkulma modları /Buckling Modes



Euler Burkulma Gerilmesi /Euler Buckling Strength

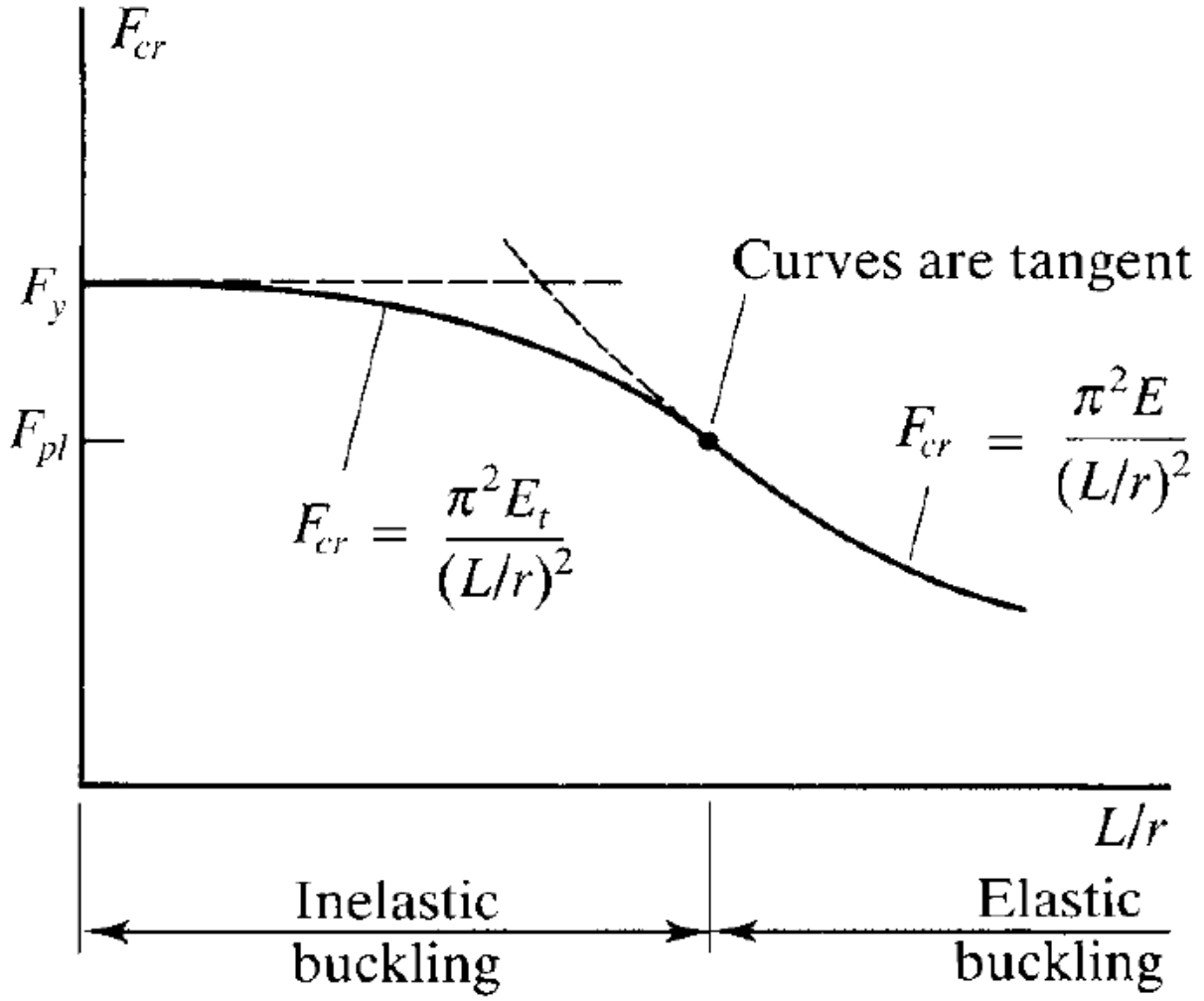
$$P_{cr,e} = \frac{\pi^2 EI}{L^2}$$

$$P_{cr} = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 EAr^2}{L^2} = \frac{\pi^2 EA}{(L/r)^2}$$

$$F_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{(L/r)^2}$$

L/r narinlik oranı
F_{cr}=burkulma gerilmesi

Elastik ve Elastik olmayan burkulma /Elastic and Inelastic Global Buckling

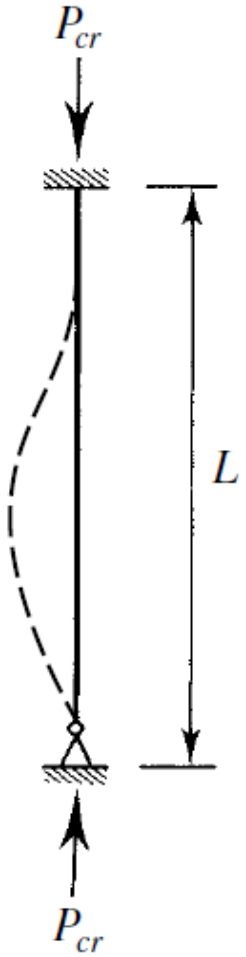


Burkulma Boyu Katsayısı /Efective Length Factor

Table C-C2.1
***K* Values for Columns**

Buckled shape of column is shown by dashed line.	(a)	(b)	(c)	(d)	(e)	(f)
						
Theoretical *K* value	0.5	0.7	1.0	1.0	2.0	2.0
Recommended design value when ideal conditions are approximated	0.65	0.80	1.2	1.0	2.10	2.0
End condition code	 *Rotation fixed and translation fixed* *Rotation free and translation fixed* *Rotation fixed and translation free* *Rotation free and translation free*					

Burkulma Boyu Katsayısı /Efective Length Factor



$$P_{cr} = \frac{2.05\pi^2 EA}{(L/r)^2} = \frac{\pi^2 EA}{(0.70L/r)^2}$$

$$P_{cr} = \frac{\pi^2 EA}{(KL/r)^2}$$

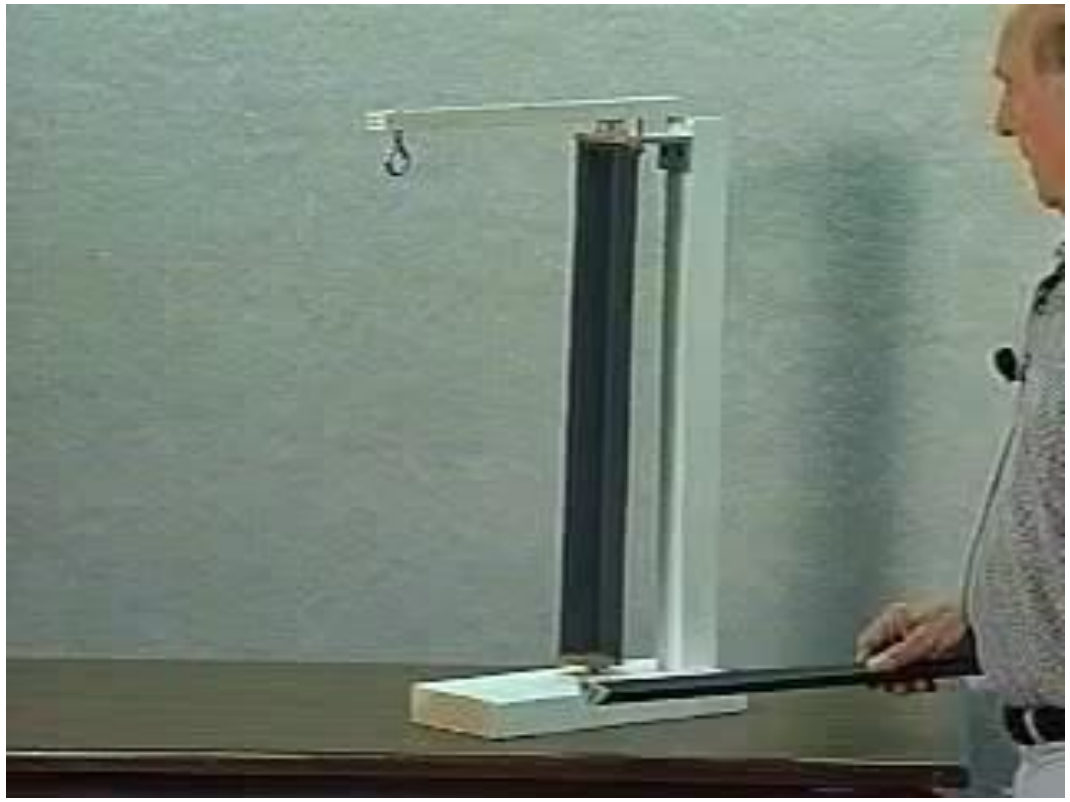
Burkulma Boyu Katsayısı /Efective Length Factor



Burkulma Boyu Katsayısı /Efective Length Factor



Burkulma Boyu Katsayısı /Efective Length Factor



Akma Sınır Durumu /Limit State of Yielding

$$P_n = F_{cr} A_g$$

For LRFD,

$$P_u \leq \phi_c P_n$$

where

P_u = sum of the factored loads

ϕ_c = resistance factor for compression = 0.90

$\phi_c P_n$ = design compressive strength

Bölüm 8. Eksenel Basınç Kuvveti Etkisi /Design of Members For Compression

Narinlik Kontrolu (5.1a ya göre)

- Başlık/lar için kontrolü (Narin Olmayan / Narin)
- Gövde için kontrol (Narin Olmayan / Narin)
  - Narin olmayan başlık ve gövde, **Narin olmayan** kesit olur. Herhangi biri narin çıkarsa kesit **narindir.**

Eksenel basınç kapasitesi

Basınç kapasitesi kritik burkulma gerilmesi ile alanın çarpımına eşittir. Eleman serbest boyu burkulma katsayısı hesabına bağlıdır. Serbest boy çarpı K(burkulma katsayısı) ile bulunur (L)

Kritik burkulma gerilmesi L/i'den hesaplanan narinliğe göre

- $L/i<4{,}71\sqrt{(E/fy)}$ ise elastik olmayan burkulma durumu geçerlidir. Fcr=0,658 … bulunur
- $L/i>4{,}71\sqrt{(E/fy)}$ ise elastik burkulma durumu geçerlidir. Fcr=0,877 … den bulunur

Bölüm 8. Eksenel Basınç Kuvveti Etkisi /Design of Members For Compression

Formüllerde bulunan Fe değeri ana değişkendir. Farklı burkulma modları için bu değerin hesaplanması gerekmektedir.

Eğilmeli burkulma durumu çift simetri eksenli **I** profillerde oluşur. **Burulmalı burkulma** durumu **+ (Haç)** profillerinde **Eğilmeli burulmalı burkulma U** , Yarım I (T Kesitler), Korniyerlerde, çift korniyerlerde

- Elastik burkulma yükü hesaplanıp Fcr hesabı yapılır.

Narin kesitler için kontrol

- Narin eleman parçasının etkin alanı bulunarak Fcr ile çarpılır.

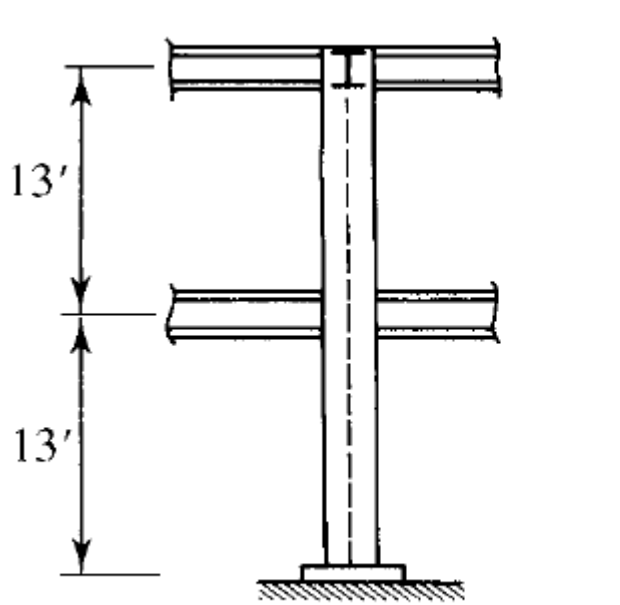
Yerel burkulma veya buruşma /Local Buckling or Shear Buckling (Wrinkling Behaviour)

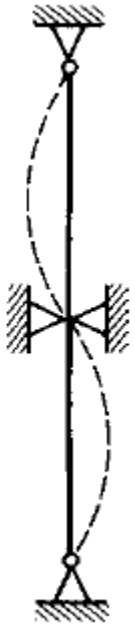




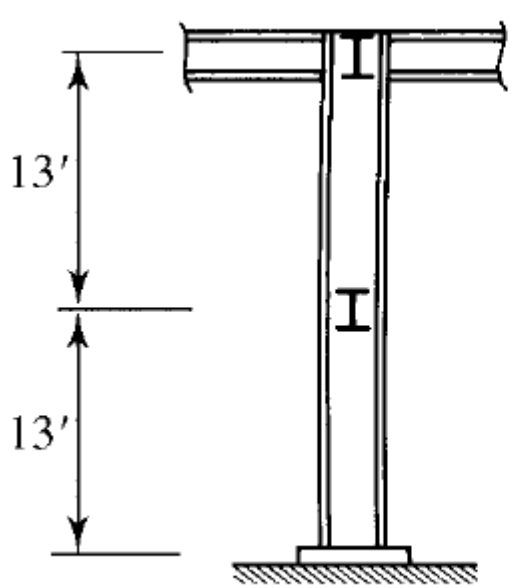


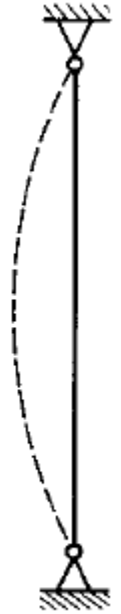
Kolonlarda burkulma /Column Buckling





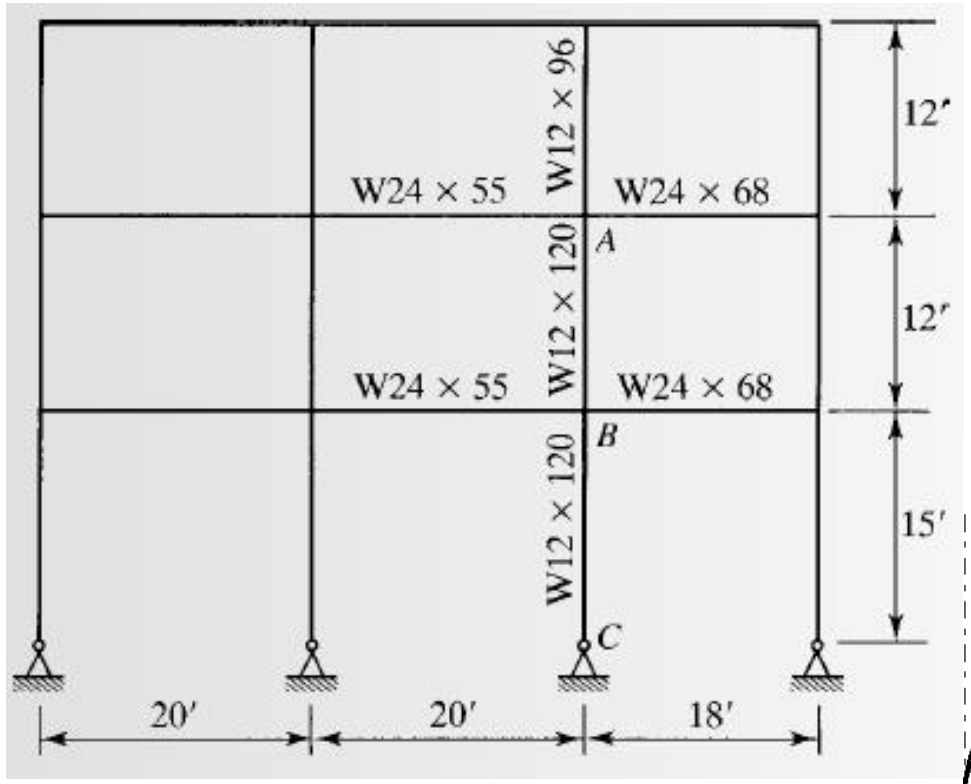
$KL = 13'$

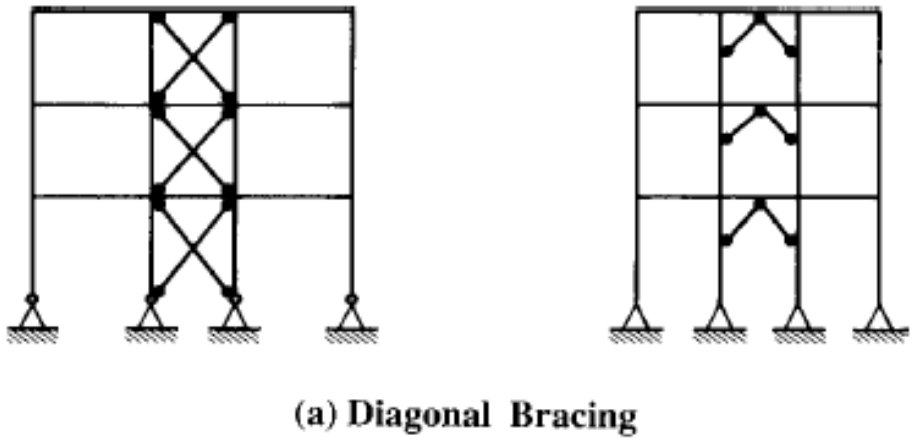




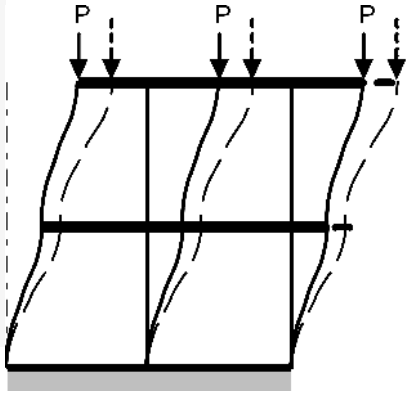
$KL = 26'$

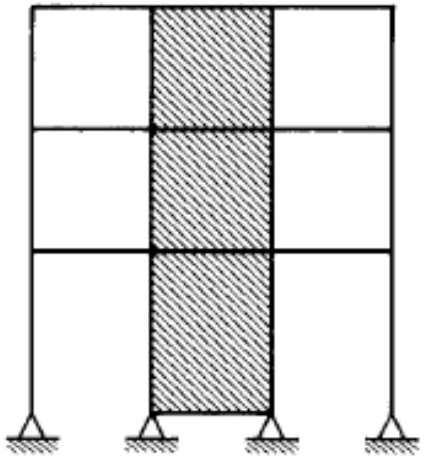
Yanal Ötelenmesi Önlenmiş veya Önlenmemiş Yapılar /Non-Sideway or Sideway Structures





(a) Diagonal Bracing





(b) Shear Walls

Basınç Çubukları İhtiva Eden Bazı Yapılar
/Some Structures Including Compression Members







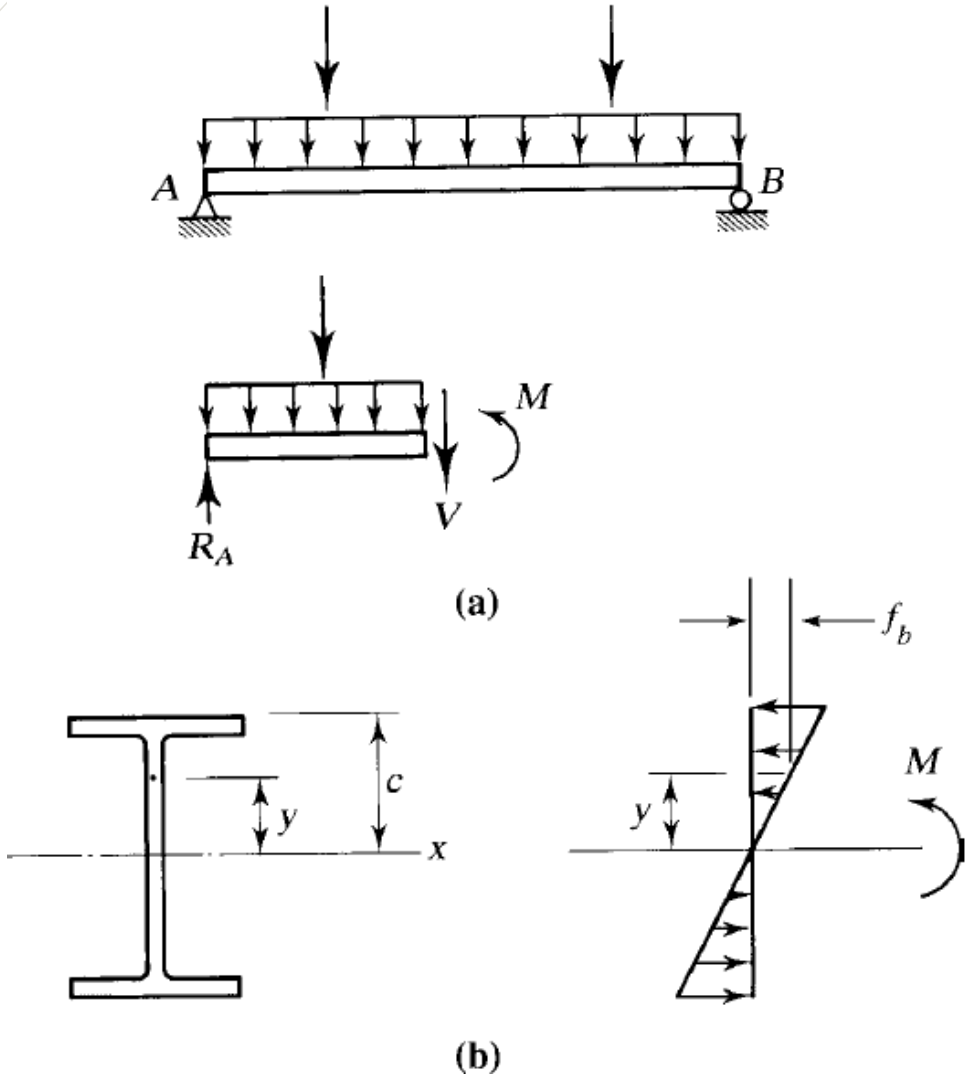




Kiriş Elemanı /Flexural Member or Beam

- Asal eksenlerinden herhangi biri eğilme etkisi altındaki elemanlara kiriş denir.
- Dayanım Hesabı yöntemi
  - Eğilme momenti etkisi
  - Kesme kuvveti etkisi

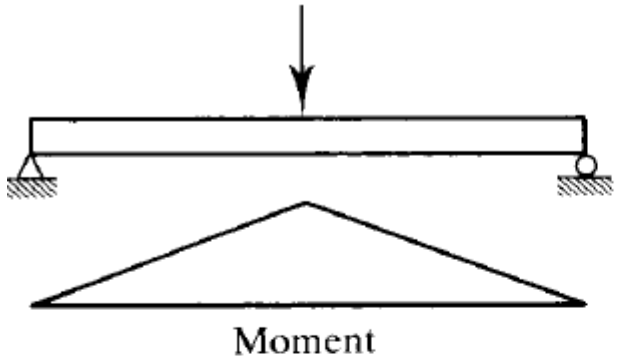
Eğilme Altında Normal Gerilme ve Plastik Moment /Normal Stress Under Bending and Plastic Moment

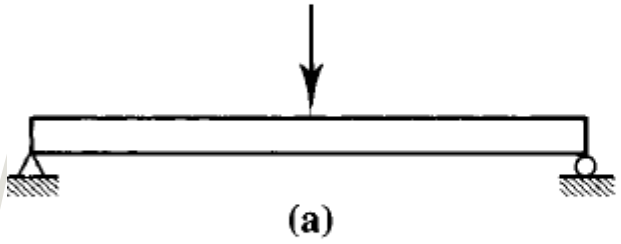


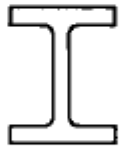
$$f_b = \frac{My}{I_x}$$

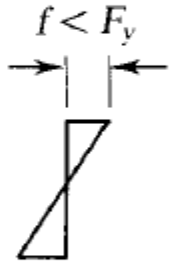
$$f_{\max} = \frac{Mc}{I_x} = \frac{M}{I_x/c} = \frac{M}{S_x}$$

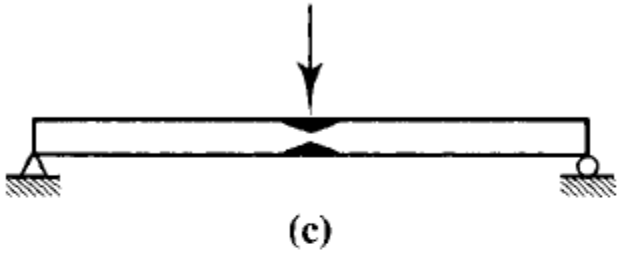
Eğilme Altında Normal Gerilme ve Plastik Moment /Normal Stress Under Bending and Plastic Moment

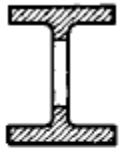


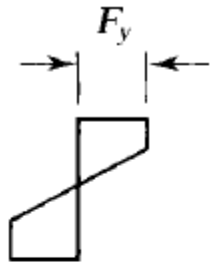


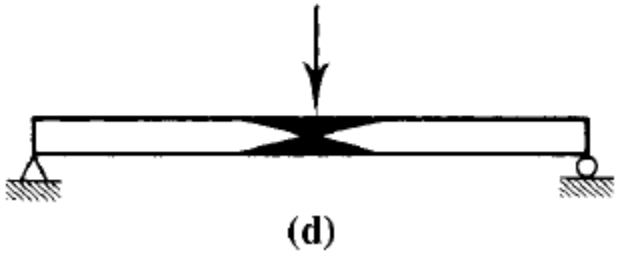




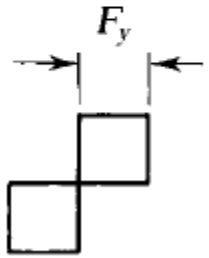


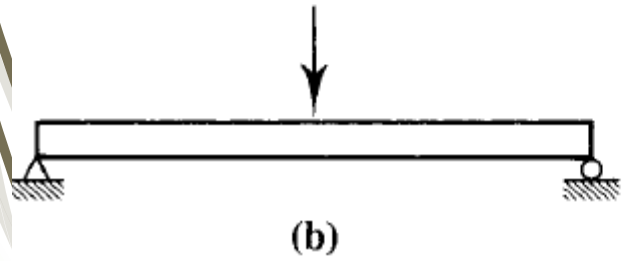




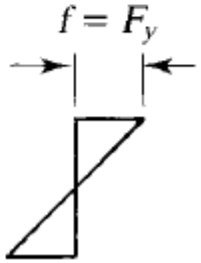






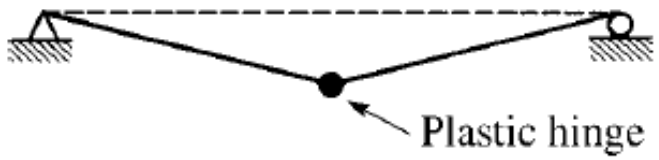


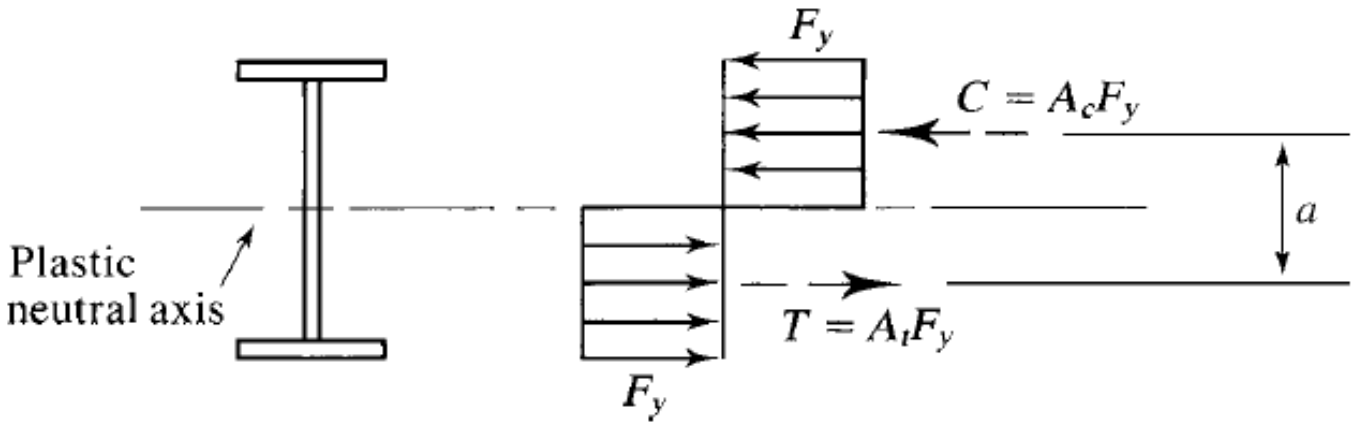




$$M_y = F_y S_x$$

Eğilme Altında Normal Gerilme ve Plastik Moment /Normal Stress Under Bending and Plastic Moment





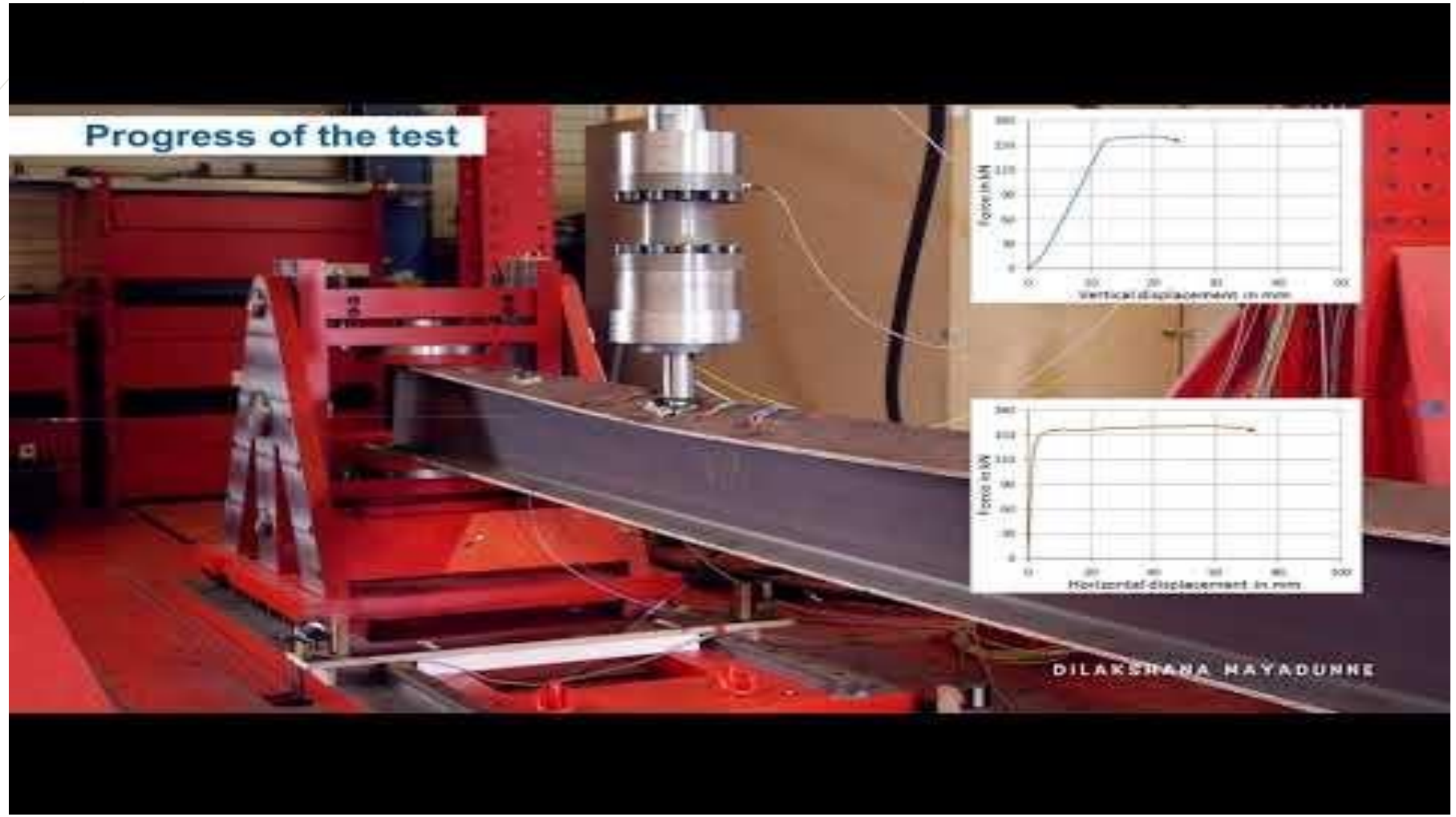
$$M_p = F_y(A_c)a = F_y(A_t)a = F_y\left(\frac{A}{2}\right)a = F_y Z$$

A = total cross-sectional area

a = distance between the centroids of the two half-areas

$Z = \left(\frac{A}{2}\right)a$ = plastic section modulus

Lateral Torsional Buckling Test



Genel ve Yerel Stabilite ile Başka Problemlerinin Taşıyıcı Sistem Davranışındaki Yeri /Global and Local Stability as well Other Failure Issues In Structure Behaviour



(a) Güney-batı cephesi



(g) Aşık bağlantı hasarı

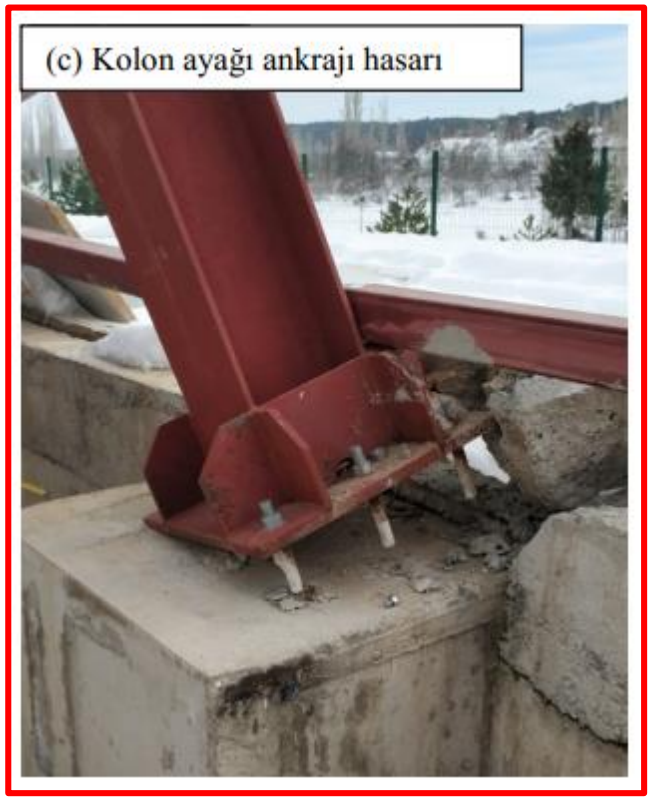


(e) Aşık bağlantı hasarı

Genel ve Yerel Stabilite ile Başka Problemlerinin Taşıyıcı Sistem Davranışındaki Yeri /Global and Local Stability as well Other Failure Issues In Structure Behaviour



(c) Kolon devrilmesi



Genel ve Yerel Stabilite ile Başka Problemlerinin Taşıyıcı Sistem Davranışındaki Yeri /Global and Local Stability as well Other Failure Issues In Structure Behaviour



(c) Kolon üstü kiriş hasarı



(d) Kolon üstü kiriş hasarı

Genel ve Yerel Stabilite ile Başka Problemlerinin Taşıyıcı Sistem Davranışındaki Yeri /Global and Local Stability as well Other Failure Issues In Structure Behaviour

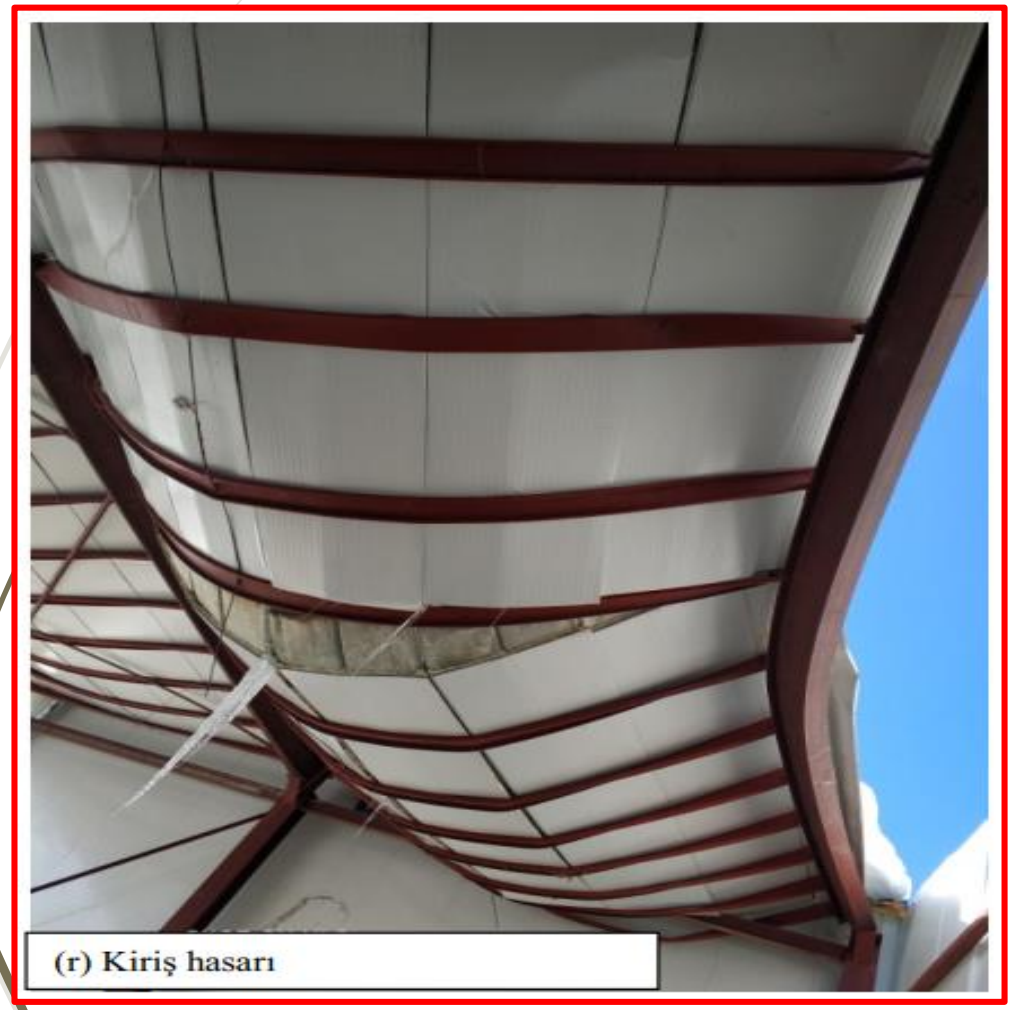


(a) Kolon üstü kiriş hasarı



(b) Kolon üstü kiriş hasarı

Genel ve Yerel Stabilite ile Başka Problemlerinin Taşıyıcı Sistem Davranışındaki Yeri /Global and Local Stability as well Other Failure Issues In Structure Behaviour

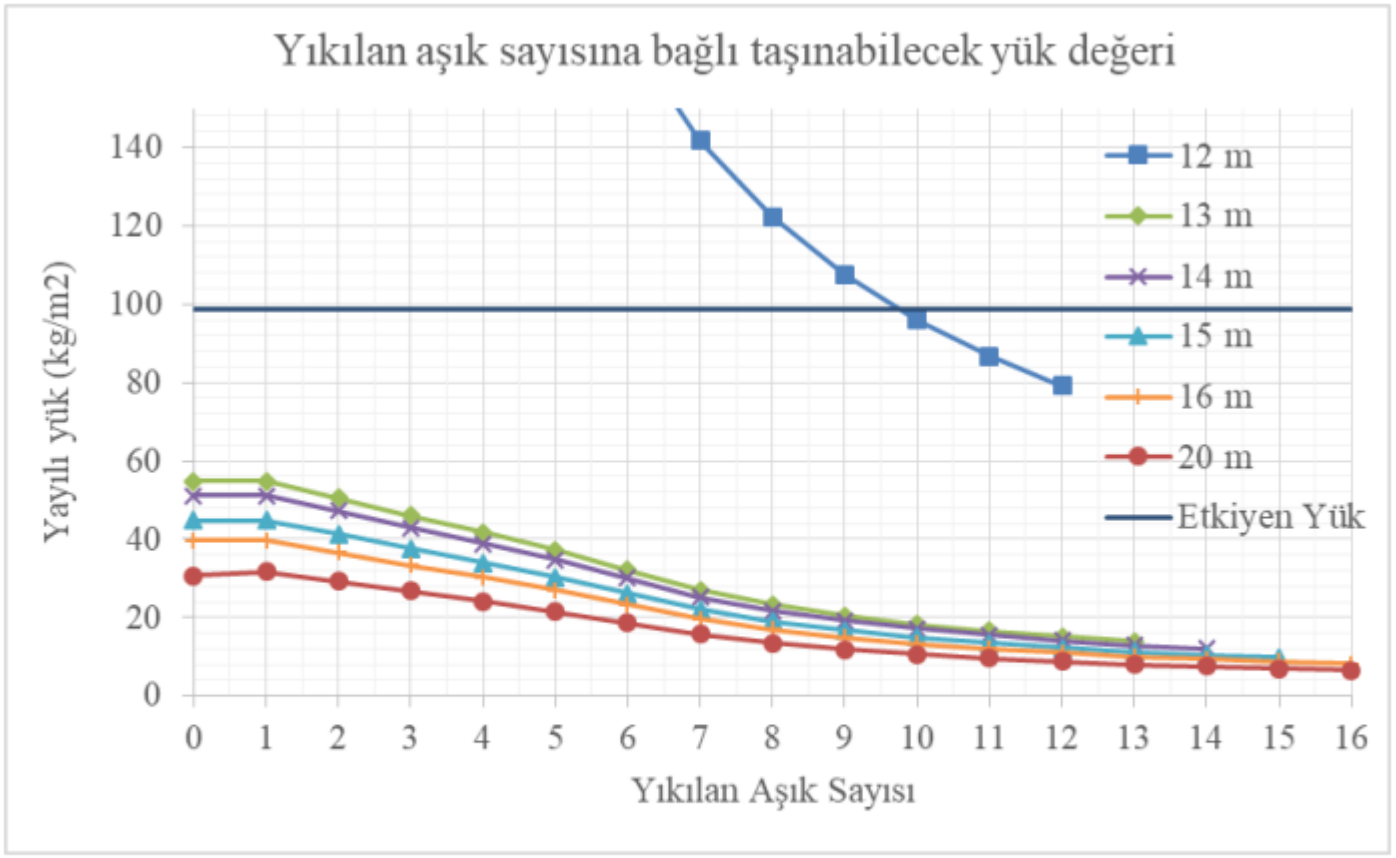


(r) Kiriş hasarı



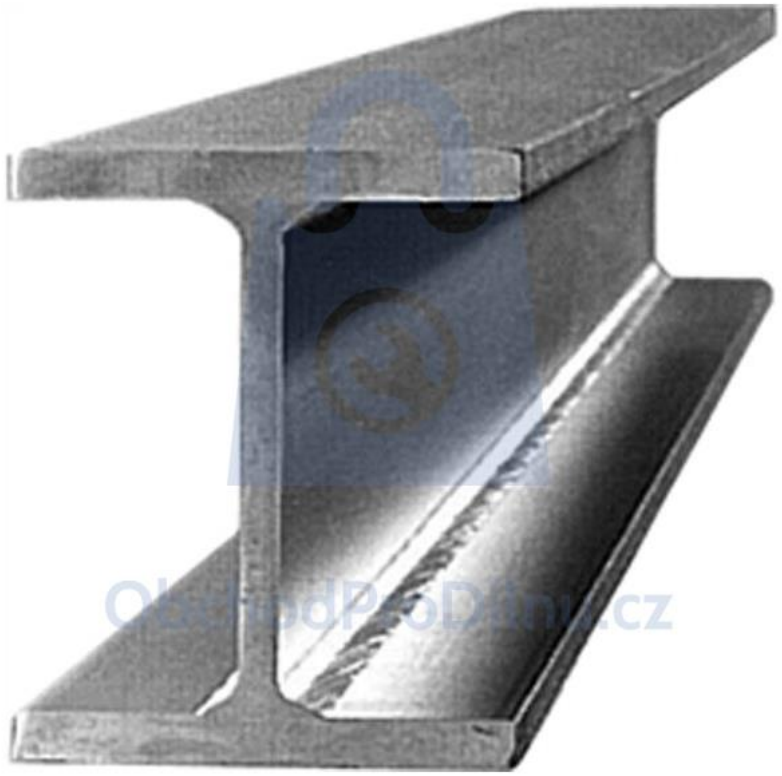
(h) Kiriş hasarı

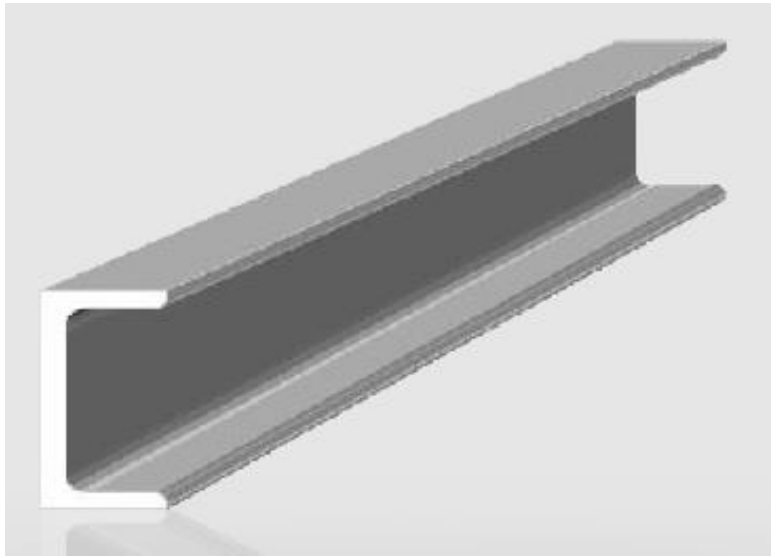
Genel ve Yerel Stabilite ile Başka Problemlerinin Taşıyıcı Sistem Davranışındaki Yeri /Global and Local Stability as well Other Failure Issues In Structure Behaviour

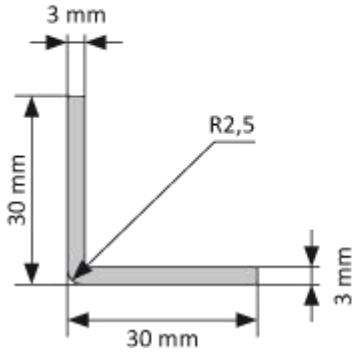


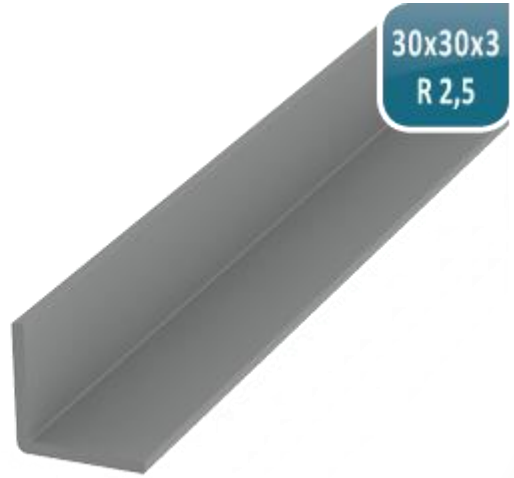
Şekil 18. Ana kirişte yıkılan aşık sayısına bağlı taşınabilecek yük değerinin farklı eleman boyları için değişim grafiği

Rijitleştirilmiş ve rijitleştirilmemiş kesit parçası /Stiffened and Unstiffened cross-sectional part

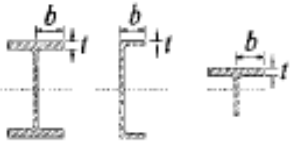
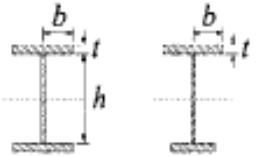
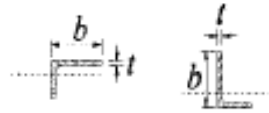
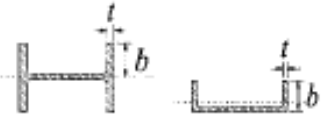


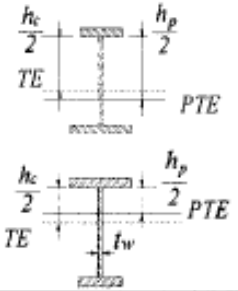
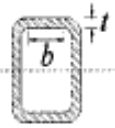
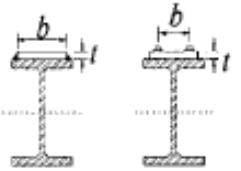
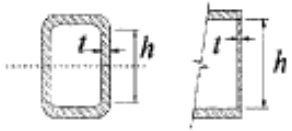
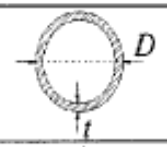






TABLO 5.1B – EĞİLME MOMENTİNİN BASINÇ BİLEŞENİ ETKİSİNDEKİ ENKESİT PARÇALARI İÇİN GENİŞLİK / KALINLIK ORANLARI

	Durum	Tanım	Genişlik/ Kalınlık Oranı, λ	Genişlik / Kalınlık Oranı Sınır Değerleri		Enkesit
				λ_p (kompakt / kompakt olmayan)	λ_r (kompakt olmayan / narin)	
Rijitleştirilmemiş Enkesit Parçası	10	Hadde I-profiller, U-profiller ve T-enkesitli elemanların başlıkları	b/t	$0.38\sqrt{\frac{E}{F_y}}$	$1.00\sqrt{\frac{E}{F_y}}$	
	11	Tek ve çift simetri eksenli yapma I-enkesitli elemanların başlıkları	b/t	$0.38\sqrt{\frac{E}{F_y}}$	$0.95\sqrt{\frac{k_c E}{F_L}}$ [a] [b]	
	12	Tek korniyerlerin kolları	b/t	$0.54\sqrt{\frac{E}{F_y}}$	$0.91\sqrt{\frac{E}{F_y}}$	
	13	Zayıf eksen etrafında eğilme etkisindeki tüm I-enkesitli elemanlar ve U-profiller	b/t	$0.38\sqrt{\frac{E}{F_y}}$	$1.00\sqrt{\frac{E}{F_y}}$	
	14	T-enkesitli elemanların gövdeleri	d/t	$0.84\sqrt{\frac{E}{F_y}}$	$1.03\sqrt{\frac{E}{F_y}}$	

	Durum	Tanım	Genişlik/ Kalınlık Oranı, λ	Genişlik / Kalınlık Oranı Sınır Değerleri		Enkesit
				λ_p (kompakt / kompakt olmayan)	λ_r (kompakt olmayan / narin)	
Rijitleştirilmiş Enkesit Parçası	15	U-profillerin ve çift simetri eksenli I-profillerin gövdeleri	h/t_w	$3.76\sqrt{\frac{E}{F_y}}$	$5.70\sqrt{\frac{E}{F_y}}$	
	16	Tek simetri eksenli I-enkesitli elemanların gövdeleri	h_c/t_w	$\frac{\frac{h_c}{h_p}\sqrt{\frac{E}{F_y}}}{\left(0.54\frac{M_p}{M_y}-0.09\right)^2} \le \lambda_r$ [c]	$5.70\sqrt{\frac{E}{F_y}}$	
	17	Üniform cidar kalınlıklı kutu enkesitli elemanların başlıkları	b/t	$1.12\sqrt{\frac{E}{F_y}}$	$1.40\sqrt{\frac{E}{F_y}}$	
	18	Takviye levhaları ve birleşim araçları arasında kalan diyafram levhaları	b/t	$1.12\sqrt{\frac{E}{F_y}}$	$1.40\sqrt{\frac{E}{F_y}}$	
	19	Kutu enkesitli elemanların gövdeleri	h/t	$2.42\sqrt{\frac{E}{F_y}}$	$5.70\sqrt{\frac{E}{F_y}}$	
	20	Boru enkesitli elemanlar	D/t	$0.07\frac{E}{F_y}$	$0.31\frac{E}{F_y}$	
	21	Yapma kutu enkesitlerin başlıkları	b/t	$1.12\sqrt{\frac{E}{F_y}}$	$1.49\sqrt{\frac{E}{F_y}}$	


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Bölüm 9. Eğilme Moment Etkisi /Design of Members For Flexure

Narinlik Kontrolu (5.1b ye göre) *(Check of slenderness)*

- Başlıklar için kontrolü (Kompakt / Kompakt olmayan / Narin) *(For check of flange, Compact/Non-compact/Slender)*
- Gövde için kontrol (Kompakt / Kompakt olmayan / Narin) *(For check of web, Compact/Non-compact/Slender)*

  Kompakt gövde ve kompakt başlık var ise kesit kompakt olur. Kesit parçalarının herhangi biri kompakt olmayan ise kesit kompakt olmayan olur. Herhangi biri narin çıkarsa kesit narindir denir. *(If there is a compact web and compact flange, the cross section is called compact. If any of the section elements are non-compact, the cross section will be non-compact. If anyone is slender, the cross section is called slender)*

Kompakt kesitler için hesap *(Design of Compact Section)*

İki farklı kapasite hesaplanır en küçüğü alınır: *(Two design strength are calculated and the smallest is taken)*

Bölüm 9. Eğilme Moment Etkisi /Design of Members For Flexure

- Akma durumu için *(For limit state of yielding)*
  - Kesit plastik mukavemet momenti akma dayanımı ile çarpılarak moment kapasite Mp bulunur. *(The nominal moment strength Mp is found by multiplying the yield strength with plastci section modulus)*
- Yanal burulmalı burkulma için *(For limit state of lateral torsional buckling)*
  - Kiriş plastik uzunluk Lp ve Elastik uzunluk Lr değerleri hesaplanıp serbest boy Lb ile karşılaştırılır. *(The limiting unbraced length for limit state of yielding, Lp and the limiting unbraced length for inealstic lateral torsional buckling, Lr are calculated and length between points that are either braced against lateral displacement of the compression flange or braced against twist of the cross section, Lb is campared with Lp and Lr)*
- Lb<Lp ise bu durum oluşmaz, kapasite hesaplanmasına gerek yoktur. *(If Lb<Lp, this limit state is not valid, therefore, any calculation accordance with this limit state is not required)*

Bölüm 9. Eğilme Moment Etkisi /Design of Members For Flexure

- Lp<Lb<Lr ise moment kapasitesi Cb değeri dikkate alınarak hesaplanır. (*If Lp<Lb<Lr, the moment capacity is calculated by considering the Cb value.)*
- Lb>Lr ise moment kapasitesi Cb değeri dikkate alınarak hesaplanır. (*If Lb>Lr, the moment capacity is calculated by considering the Cb value.*)
  - Hesaplanan Moment kapasiteleri hiçbir durumda **Mp den büyük olamaz**. (*The calculated moment capacities shall in no case be greater than Mp (plastic moment capacities).*

Kompakt olmayan kesitler için hesap (*Calculation for non-compact sections)*

- Yanal burulmalı burkulma için (*For lateral torsional buckling)*
- Kiriş plastik uzunluk Lp ve Elastik uzunluk Lr değerleri hesaplanıp serbest boy Lb ile karşılaştırılır (*Beam plastic length Lp and Elastic length Lr values are calculated and compared with free length Lb.*

Bölüm 9. Eğilme Moment Etkisi /Design of Members For Flexure

- Lb<Lp ise bu durum oluşmaz kapasite hesaplanmasına gerek yoktur. *(If Lb<Lp, this situation does not occur. There is no need to calculate the capacity.)*
- Lp<Lb<Lr ise moment kapasitesi Cb değeri dikkate alınarak hesaplanır *(If Lp<Lb<Lr, the moment capacity is calculated by considering the Cb value.)*
- Lb>Lr ise moment kapasitesi Cb değeri dikkate alınarak hesaplanır *(If Lb>Lr, the moment capacity is calculated by considering the Cb value.*
  - Hesaplanan Moment kapasiteleri hiçbir durumda Mp den büyük olamaz. *(The calculated moment capacities shall in no case be greater than Mp.)*

Başlıkta yerel burkulma durumu için *(For local buckling in the Flange)*

- Kompakt olmayan veya narin olması durumları için hesaplanır. (*Calculated for non-compact or delicate situations.)*

Basınç başlığında akma durumu için *(For yielding limit state at the compression flange)*

Çekme başlığı akma durumu için *(For yielding limit state at the tension flange)*

Bölüm 9. Eğilme Moment Etkisi /Design of Members For Flexure

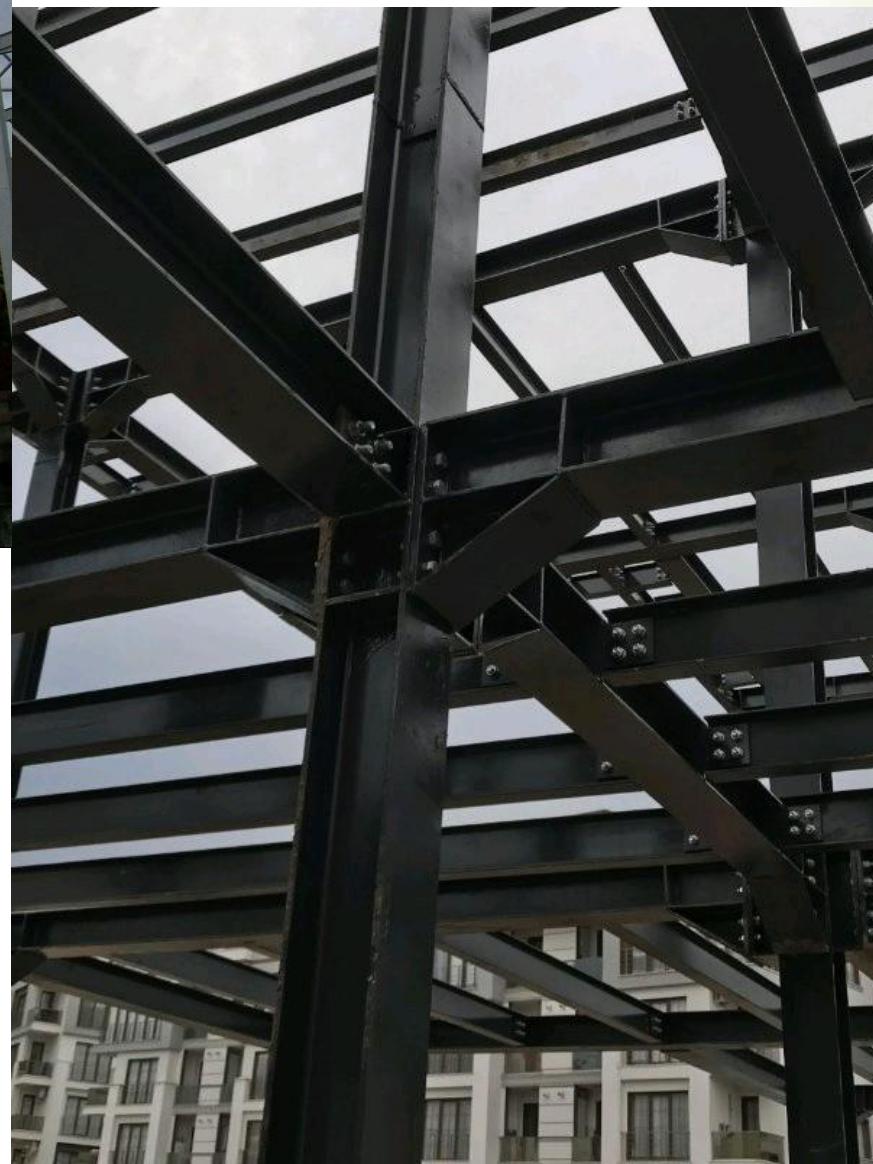
- Çekme başlığı elastik mukavemet momenti basınç başlığından büyükse bu durum kontrol edilmesine gerek yoktur. *(If the tension flange elastic moment of strength is greater than the compression flange, this situation does not need to be checked.)*
- Çekme başlığı akma sınır durumu plastikleşme katsayısı ile çekme başlığı akma momenti çarpılarak bulunur. *(The tension flange yield limit state is found by multiplying the plasticization coefficient by the tension flange yield moment.)*

Zayıf eksenleri doğrultusunda kesitler için hesap *(Calculation for sections along weak axes)*

- Akma durumu için *(For the yield limit state)*
  - Plastik mukavemet momentinin akma dayanımı ile çarpılmasına eşittir. Bu değer elastik mukavemet momenti ile akma dayanımının 1.6 katından daha büyük olamaz. *(It is equal to plastic section modulus multiplied by the yield strength. This value cannot be greater than 1.6 times the elastic strength moment and yield strength.)*
- Yerel burkulma durumu için *(For local buckling limit state)*
  - Başlıklar kompakt ise bu durum dikkate alınmaz *(This is not taken into account if the flange are compact.)*
  - Başlık parçalarının Kompakt olmayan olması durumunda *(In case the flange parts are Non-Compact)*
    - Moment kapasitesi narinlik değeri enterpolasyanuna bağlı olarak bulunur. (*The moment capacity is found depending on the slenderness value interpolation.)*
  - Başlık parçalarının Narin olması durumunda *(In case the flange parts are slender)*
    - Moment kapasitesi Fcr ile elastik mukavemet momenti çarpılarak bulunur. *(The moment capacity is found by multiplying Fcr with the moment of elastic strength.)*







Seismic Brace Connection



Shear Connection





HCS Moment Connection



Moment Connection

Bölüm 11. Birleşik Etkiler /Design of Members For Combined Forces and Torsion

Sistem narinliği kontrolü yapılır: *(Check of slenderness is made)*

- Sistemde çapraz olan katlar yatay olarak ötelemesi yapamaz. *(Storey including brace members is considered as non-sideway)*
- Sistemde çapraz olan katlar yatay olarak öteleme yapabilir. *(Storey excluding brace members is considered as sideway)*

- İkinci mertebe teorisine göre B1 ve B2 katsayısı hesaplanarak elemanlara gelen artırılmış yük ve moment değerleri bulunur. *(Accordance with the second order theory, the B1 and B2 factors are calculated and the increased load and moment values of the elements are calculated)*
- Elemana etkiyen yük durumuna göre eksenel yük ve moment kapasiteleri önceki konulara göre bulunur. *(The axial load and moment capacities are found according to the previous topics according to the load condition affecting the member)*

Bölüm 11. Birleşik Etkiler /Design of Members For Combined Forces and Torsion

- Elemana eksenel etkiyen yük eksenel kapasiteye oranı %20 den küçük olursa moment etkin olur. Bu durumda gelen eksenel kuvvetin eksenel kapasiteye oranının yarısına gelen momentlerin moment kapasitelerine oranına eklenerek bulunur, toplam maksimum bir (1) olabilir. *(If the ratio of the axially loading to the axial capacity is less than 20%, the moment drives the total capacity. In this case, the half of the ratio of the axial force to axial capacity is added the ratios of moments to moment capacities)*
- %20 den büyük olursa eksenel yük etkin şekilde davranış gösterir. Eksenel yük etkin şekilde davranışta gelen eksenel kuvvetin eksenel kapasiteye oranına gelen momentlerin moment kapasitelerine oranının 8/9 katı eklenerek bulunan toplam maksimum bir (1) olabilir. *(If the ratio of the axially loading to the axial capacity is large than 20%, the axial force is getting significance. In this case, the ratio of the axial force to axial capacity is adede eight over nine times the ratios of moments to moment capacities)*

Birleşim elemanları /Fasteners

- Perçin
  Rivet





- Bulon (cıvata)
  Bolt

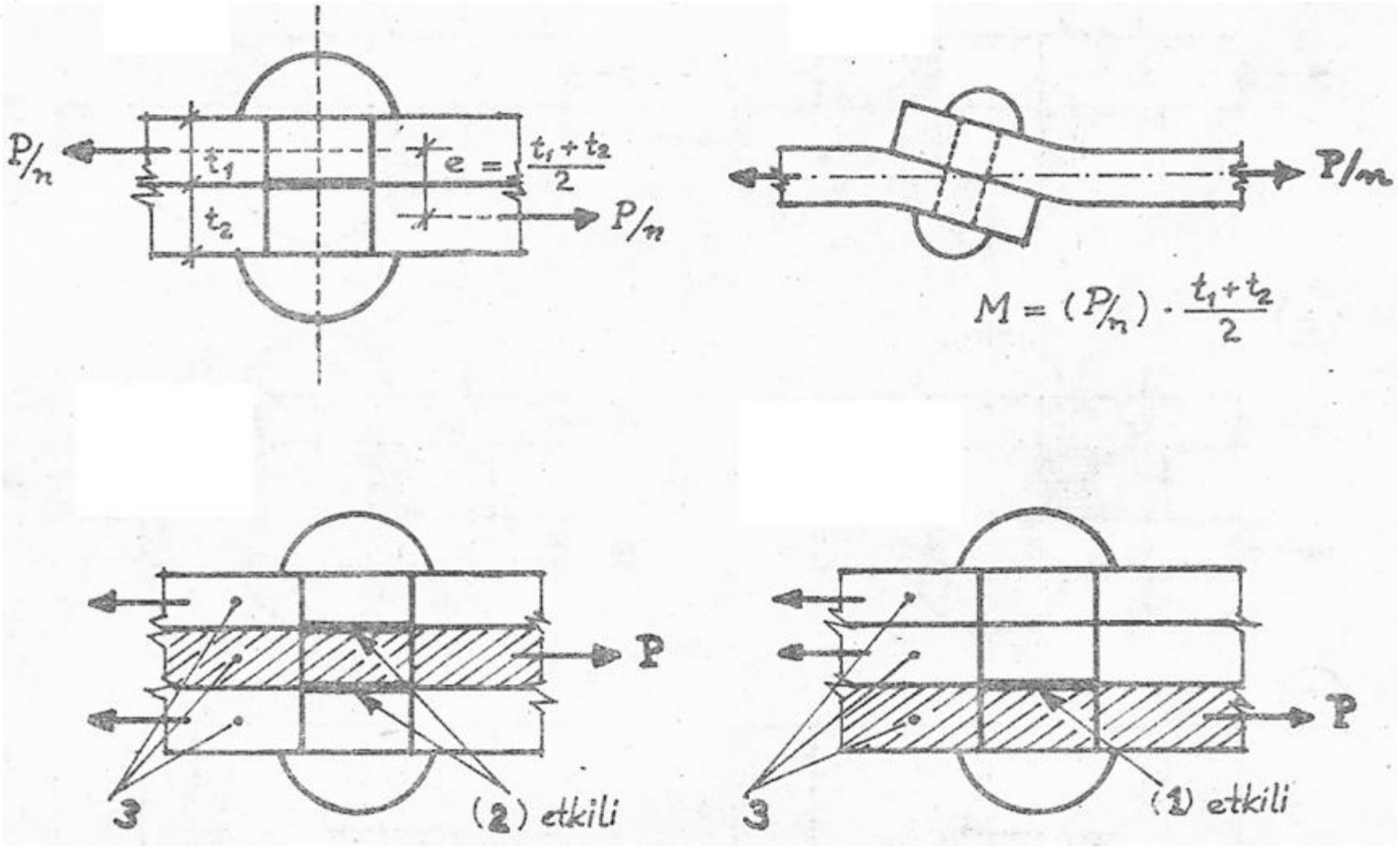


- Kaynak
  Weld





Tesir sayısı /Number of Slip Plane

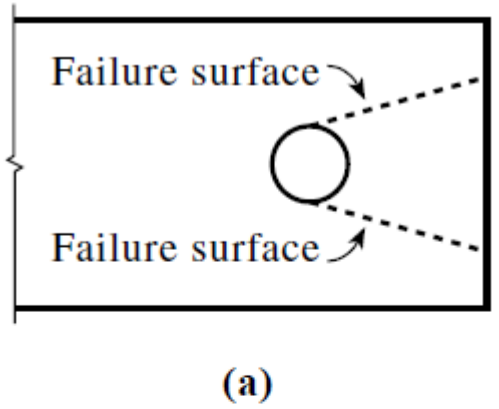


Perçin ve Bulon için emniyet gerilmeleri /Rivet or Bolt Strength

$$\frac{R_n}{2} = 0.6F_u \ell_c t$$

where

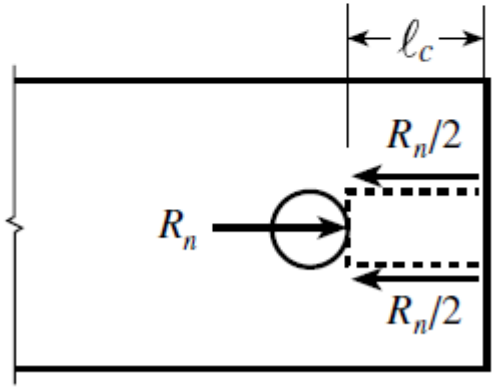
$0.6F_u$ = shear fracture stress of the connected part
ℓ_c = distance from edge of hole to edge of connected part
t = thickness of connected part



(a)

$$R_n = 1.2\ell_c t F_u \le 2.4 dt F_u$$

ℓ_c = clear distance, in the direction parallel to the applied load, from the edge of the bolt hole to the edge of the adjacent hole or to the edge of the material
t = thickness of the connected part
F_u = ultimate tensile stress of the connected part (*not* the bolt)



(b)

Perçin ve Bulon için emniyet gerilmeleri /Rivet or Bolt Strength

Bulon sınıfı	**4.6**	**4.8**	**5.6**	**5.8**	**6.8**	**8.8**	**10.9**
F_{yb}	240	320	300	400	480	640	900
F_{ub}	400	400	500	500	600	800	1000

Kaynaklı birleşimler /Welded Connection

- Kaynak, malzemelerin ısı ve/veya basınç uygulanarak birleştirilmesine denir. *(Welding is to join two or more parts which are fused together by means of heat, pressure or both forming a join as the parts cool.)*
- Bazı kaynak yöntemlerinde kaynak elektrotu, kaynak teli gibi ilaveler kullanılır. *(In some welding methods, auxiliary materials such as welding electrode and welding wire are used.)*
- Kaynak işlemi koruyucu gaz, kaynak tozu yada pasta gibi yardımcı malzemeler ile kolaylaştırılabilir. Kaynak için gerekli enerji birleşime dışarıdan uygulanır. *(Welding process can be facilitated by auxiliary materials such as shielding gas, flux or paste. The energy required for welding is applied externally to the joint.)*
- Kaynaklı birleşimler, hasar vermeden çözülemeyen tip birleşimlerdir. *(Welded joints are type of joints that cannot be broken into pieces without damaging them.)*

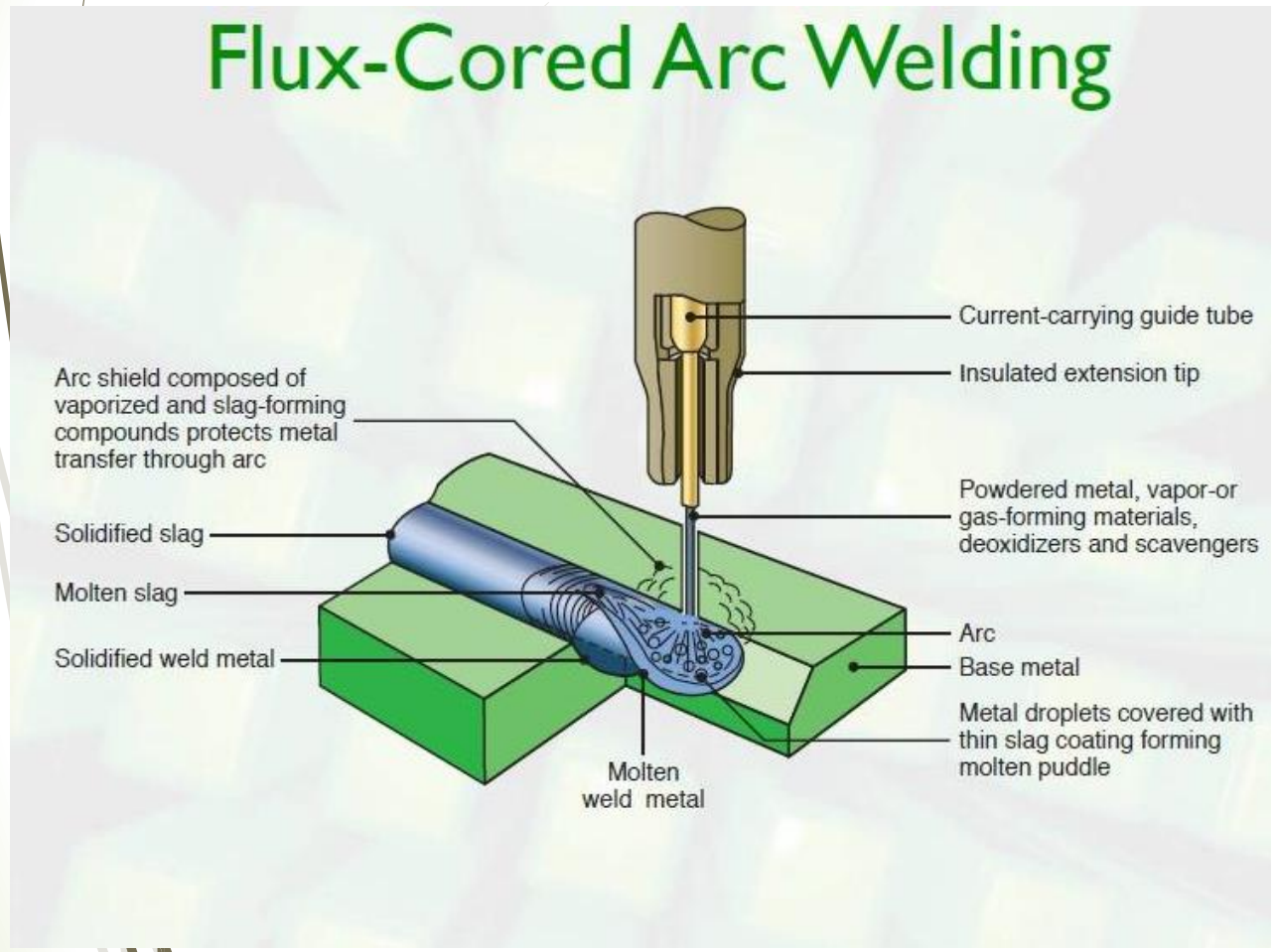
Kaynak Makineleri Türleri /Types of Welding Machine

Ergitme Kaynakları

- Elektrik arkı kaynakları *FCAW (flux cored arc welding)*
  - Standart elektrik arkı kaynağı (elektrod kaynağı) *SMAW (shielded metal arc welding)*
  - Özlü elektrodlu elektrik arkı kaynağı
  - Toz altı elektrik arkı kaynağı *SAW (submerged arc welding)*
  - Gaz altı elektrik arkı kaynağı *GMAW (gas metal arc welding)*
    - WIG (*wolfram inert gas*) kaynağı
    - MIG (*metal inert gas*) kaynağı
    - MAG (*metal active gas*) kaynağı
- Gaz Kaynağı
- Lazer Işın Kaynağı
- Elektron Işın Kaynağı

Youtube Video: Sparker Electric by Tozarc, https://www.youtube.com/channel/UC0oOCsZ7jt0wGJZJwxbDVdA/videos

Elektrik Arkı Kaynağı /Flux Core Arc Welding





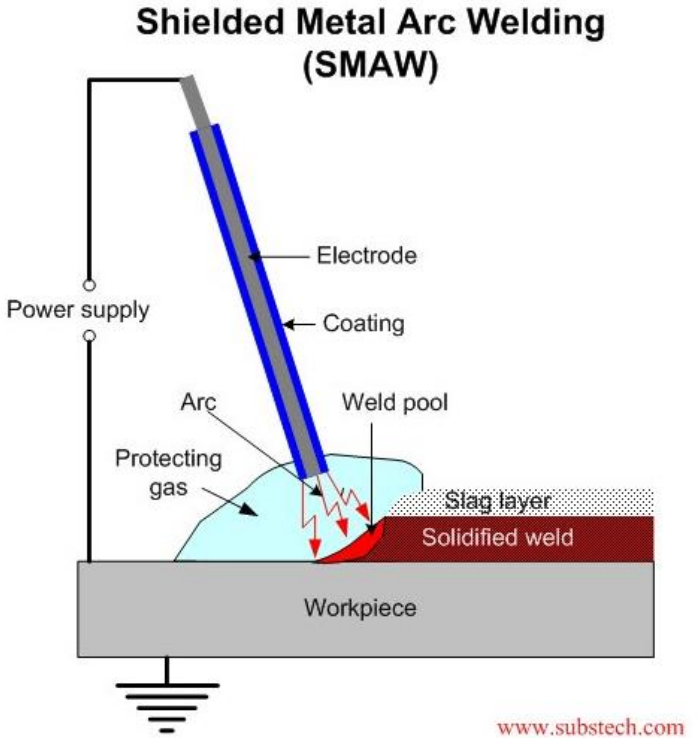
Flux-Cored Welding: The Basics for Mild Steel, https://www.millerwelds.com/resources/article-library/flux-cored-welding-the-basics-for-mild-steel

What is flux-cored arc welding (Fcaw), https://www.twi-global.com/technical-knowledge/faqs/flux-cored-arc-welding-fcaw

Standart Elektrik Arkı Kaynağı(Elektrod Kaynağı)/Shield Metal Arc Welding





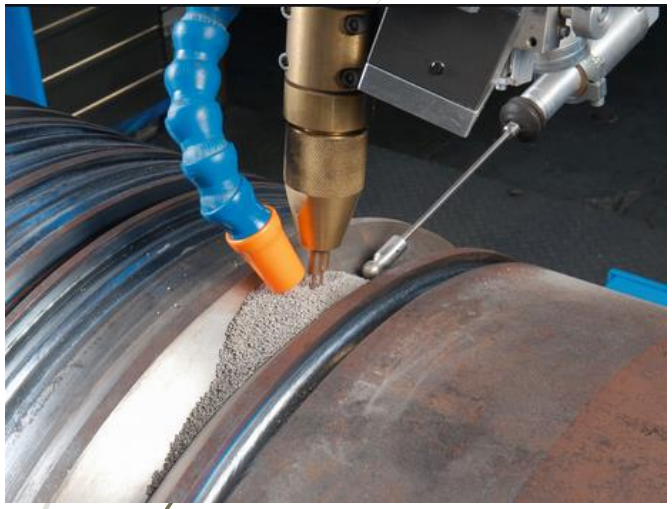


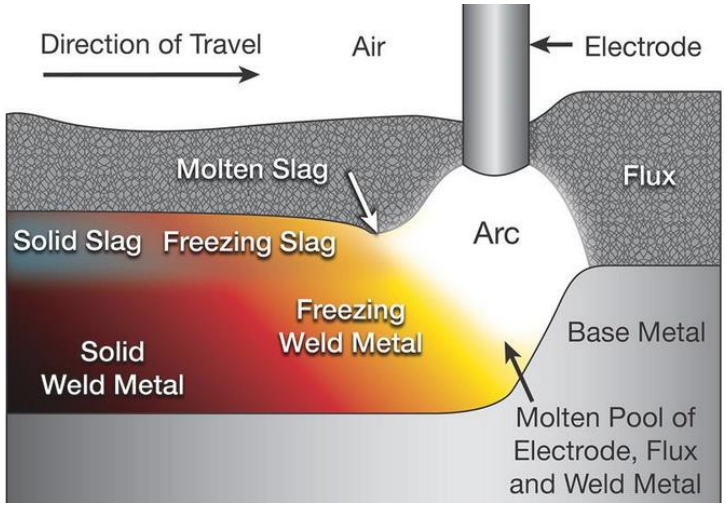


Typical Current (Amperage) Ranges				
S.N.	Electrode Diameter (mm)	Electrode		
		E6010/E6011	E6013	E7018
1	2.5	40 to 80	45 to 90	70 to 110
2	3.2	75 to 125	80 to 130	105 to 155
3	4.0	110 to 170	105 to 180	130 to 200
4	5.0	140 to 215	150 to 230	200 to 275

Reference: Shielded Metal Arc Welding, https://www.weldingandndt.com/shielded-metal-arc-welding-smaw/

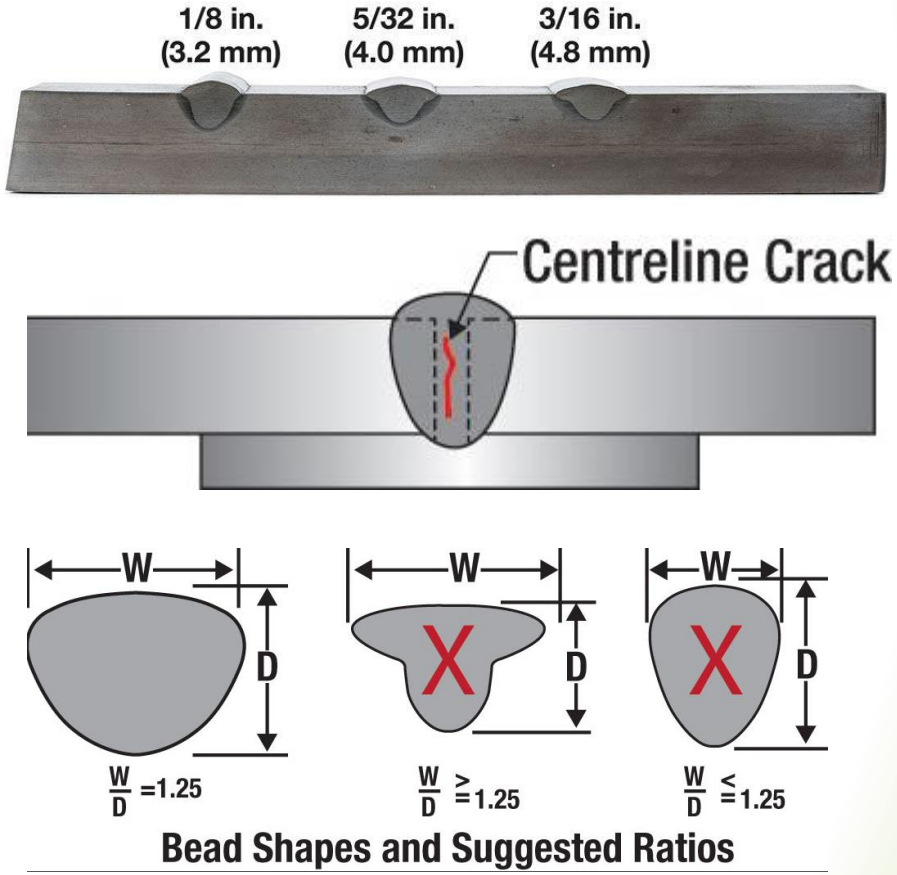
Toz Altı Elektrik Arkı Kaynağı /Submerged Arc Welding



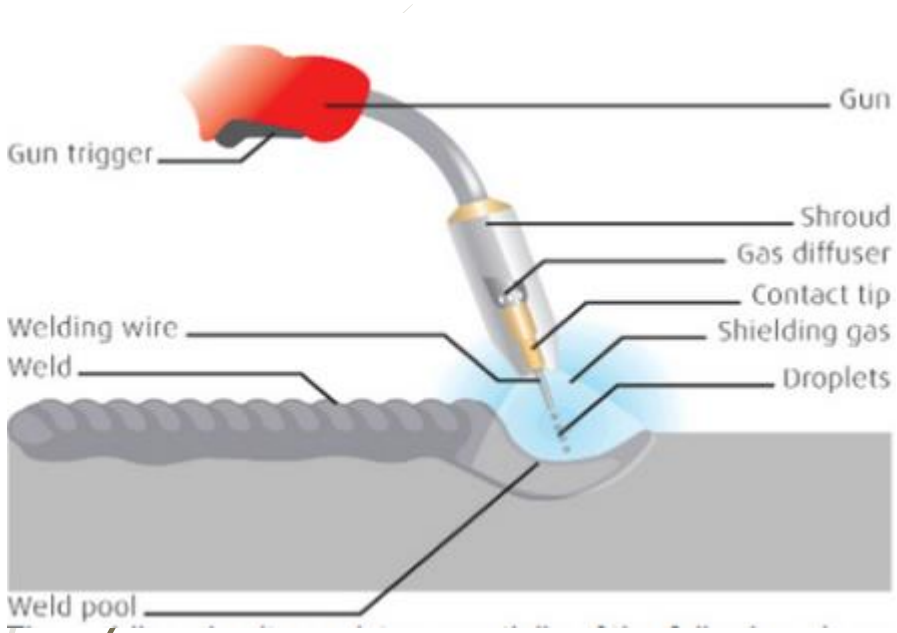


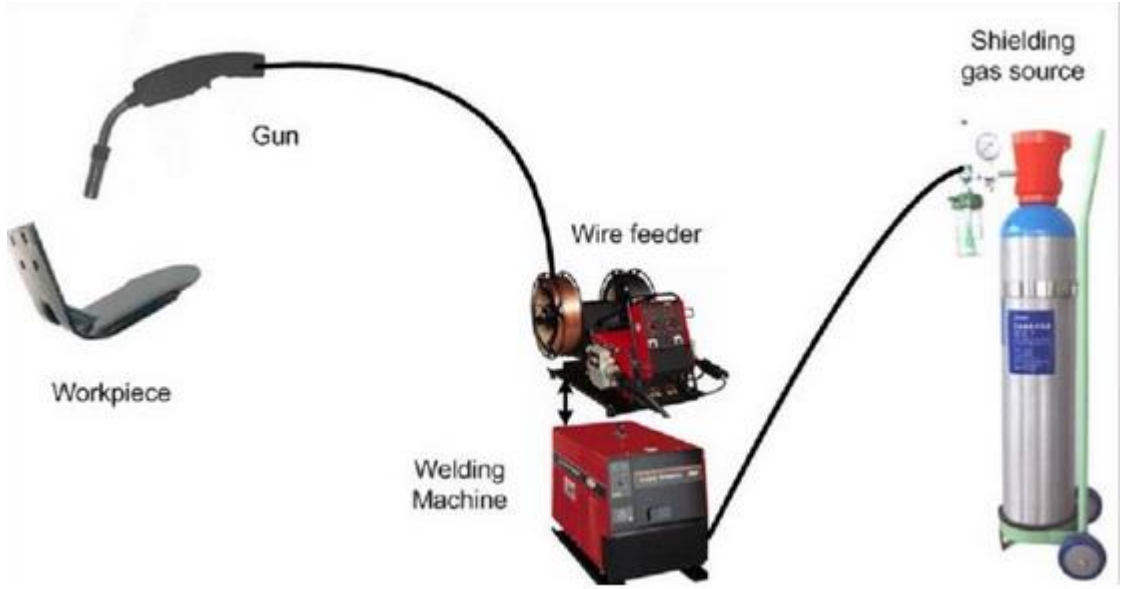


Reference: Submerged arc tech tips, https://www.canadianmetalworking.com/canadianfabricatingandwelding/article/welding/submerged-arc-tech-tips



Gaz Altı Elektrik Arkı Kaynağı /Gas Metal Arc Welding





Reference: What Is Mıg Weldıng /Gas Metal Arc Welding, https://www.sparkerweld.com/en/technology-detail/what-is-mig-welding--gas-metal-arc-welding/21

Gas metal arc welding (GMAW), sometimes referred to by its subtypes metal inert gas **MIG welding** or metal active gas (MAG) welding is an arc welding process that joins metals together by heating them with an electric arc that is established between a consumable electrode (wire) and the workpiece. A supplied gas or gas mixture acts to shield the arc and molten weld pool .

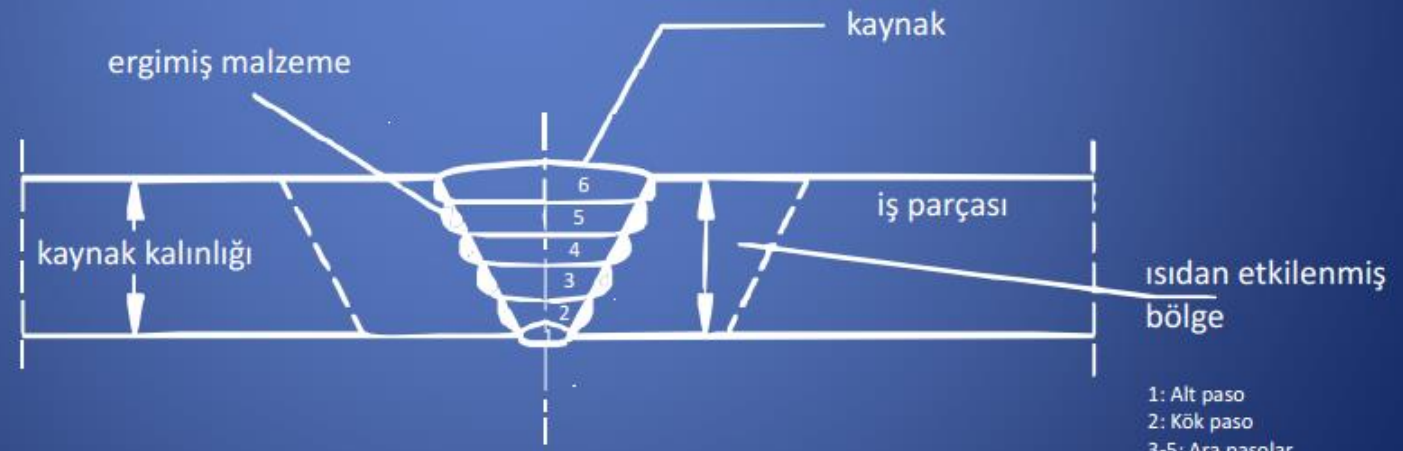
Kaynak çeşitleri /Types of Welds

Kaynak Dikişleri

Ergitme kaynağı yöntemleriyle çekilen kaynak dikişleri başlıca iki çeşittir:

Küt kaynak dikişleri: Ayni düzlemde bulunan iki levhanın yan yana getirilen kenarları boyunca çekilen kaynak dikişleri küt kaynak dikişleridir. Levha kenarlarının işleniş biçimlerine göre özel isimler alırlar.

Köşe kaynak dikişleri: İki çelik elemanın birbirine dik veya en az 60 teşkil eden yüzeyleri arasındaki köşelere çekilen dikişlere köşe kaynak dikişleri denir.



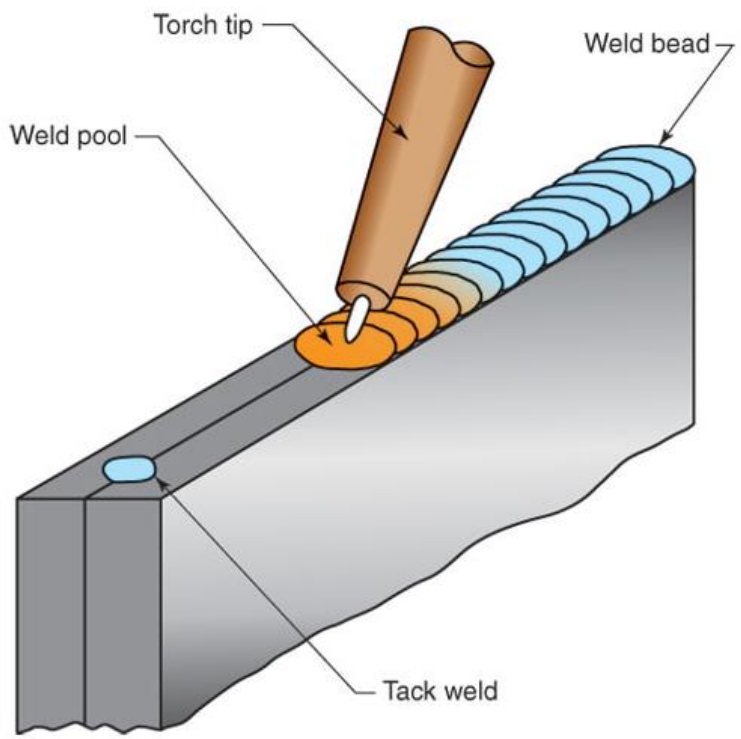
Dikiş türleri /Types of Groove or Fillet Weld

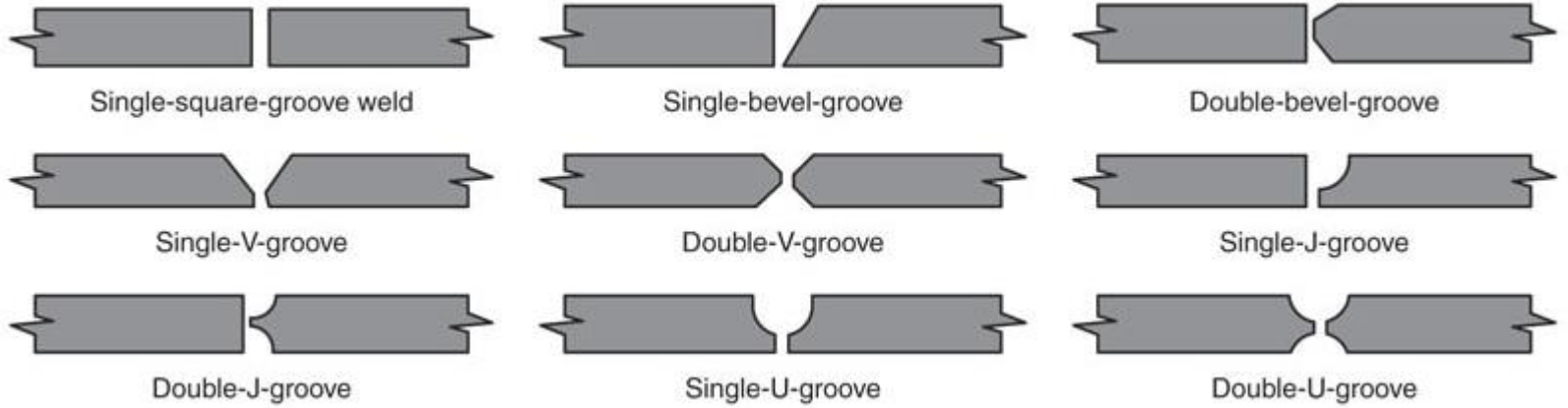
Dikiş türleri	Gösterim	Sembol
V – Dikişi		V
Yarım V – Dikişi		V
Çift V – Dikişi		X
Çift Yarım V – Dikişi (K)		K
Y – Dikişi		Y
Yarım Y – Dikişi		Y
Çift Y – Dikişi		X
Çift Yarım Y – Dikişi		K
Köşe Kaynak Dikişi		△
Çift Köşe Kaynak Dikişi		▷◁

Tam Penetras-yonlu ve **Kısmı Penetras-yonlu** → Küt Kaynak */Groove Weld*

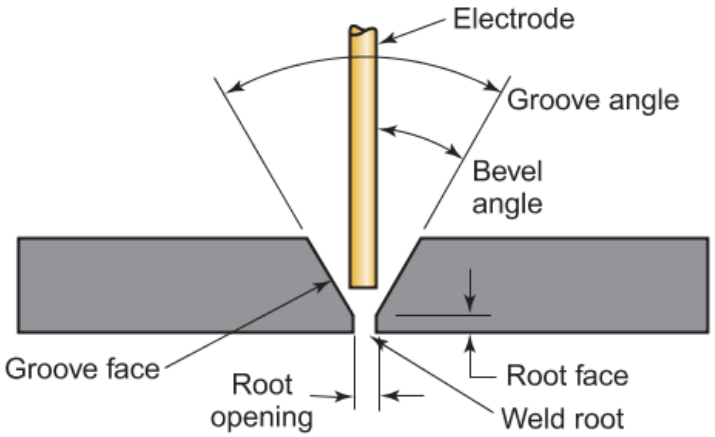
→ Köşe Kaynak */Fillet Weld*

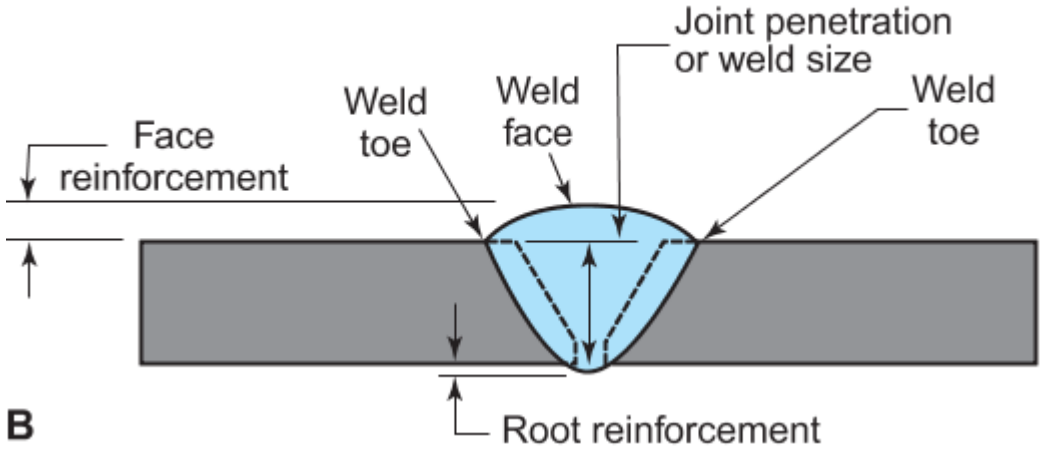
Dikiş türleri /Types of Groove or Fillet Weld



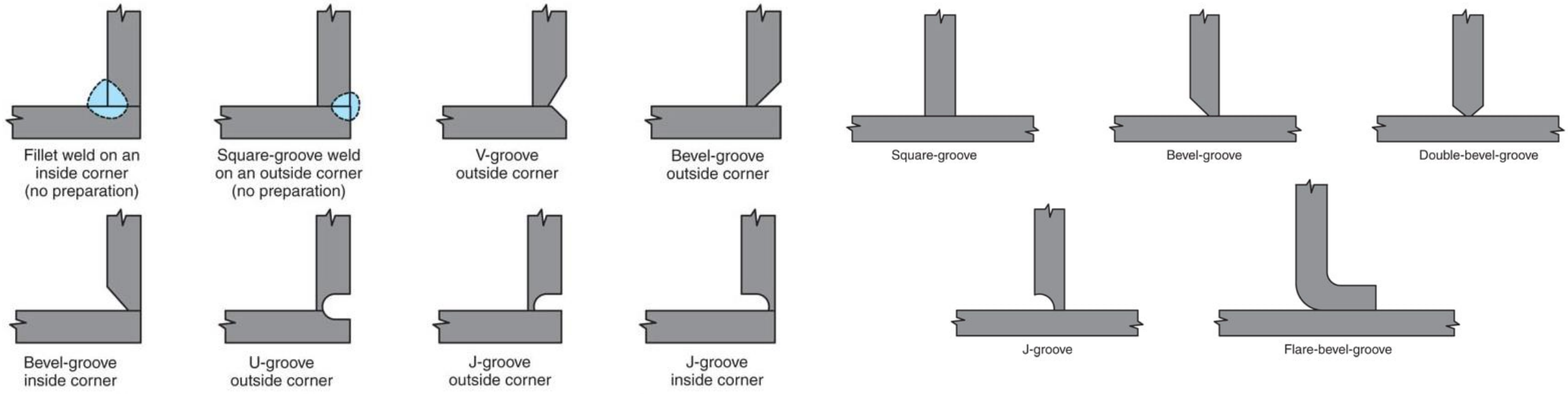


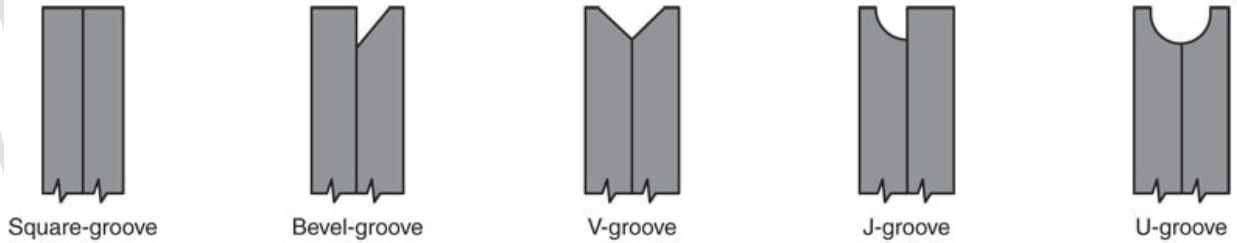
Reference:
http://www.halversoncts.com/32-types-of-welds.html

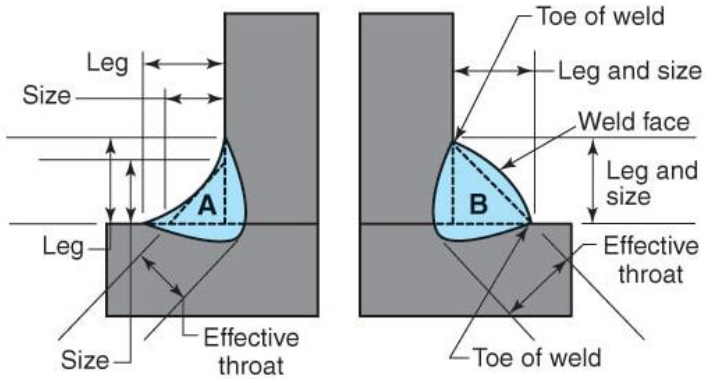




Dikiş türleri /Types of Groove or Fillet Weld







Reference: http://www.halversoncts.com/32-types-of-welds.html

Çelik malzemesine göre kaynak çeşitleri /Types of Welds Accordance With Steel Material

Base Metal		Matching Filler Metal
A36 ≤ 3/4 in. thick		60 & 70 ksi filler metal
A36 > 3/4 in. A588* A1011	A572 (Gr. 50 & 55) A913 (Gr. 50) A992 A1018	SMAW: E7015, E7016, E7018, E7028 Other processes: 70 ksi filler metal
A913	(Gr. 60 & 65)	80 ksi filler metal

*For corrosion resistance and color similar to the base metal, see AWS D1.1/D1.1M, subclause 3.7.3.

Notes:
Filler metals shall meet the requirements of AWS A5.1, A5.5, A5.17, A5.18, A5.20, A5.23, A5.28 or A5.29.
In joints with base metals of different strengths, use either a filler metal that matches the higher strength base metal or a filler metal that matches the lower strength and produces a low hydrogen deposit.

Kaynak emniyet gerilmeleri /Weld Strength

Table 8–3
Electrode Strength Coefficient, C_1

Electrode	F_{EXX} (ksi)	C1
E60	60	0.857
E70	70	1.00
E80	80	1.03
E90	90	1.16
E100	100	1.21
E110	110	1.34

Bağlantı noktaları /Connections or Junction

